SGA 5 – Batch 028

Lefschetz formula and local terms; J-adic projective systems; beginning of ℓ-adic cohomology

Working English LaTeX translation

Exposé III B. Calculation of local terms – continuation

5.11. Twisted traces

Let A be a K-algebra, let $E \in \operatorname{ob} D^-(A)$, and let $\rho : A \to A$ be an endomorphism of K-algebras. Denote by A_ρ the ring A, regarded as a left A^e-module through

$$a \otimes a' \longmapsto \rho(a)a'.$$

There is a canonical isomorphism

$$\operatorname{Hom}_A(E, A_\rho \otimes_A E) \xrightarrow{\sim} \operatorname{Hom}_A(E, E),$$

where the term on the right means the set of left A-module homomorphisms $E \to E$ which are ρ-linear, that is, such that

$$u(ax) = \rho(a)u(x).$$

For $u \in \operatorname{Hom}_A(E, A_\rho \otimes_A E)$ we shall sometimes write $\operatorname{Tr}^\rho_A(u)$, or simply $\operatorname{Tr}_\rho(u)$, for the element $\operatorname{Tr}_A(u) \in A_K$.

When ρ is an inner automorphism, say $\rho(a) = gag^{-1}$, one checks that $\operatorname{Tr}_\rho(u)$ depends only on the class of g in A_K^*, and that for a ρ-linear u one has

$$\operatorname{Tr}_\rho(u) = \text{the class of } g \text{ in } A_K \cdot \operatorname{Tr}(g^{-1}u), \tag{5.13.1}$$

where $g^{-1}u$ is the A-linear endomorphism $x \mapsto g^{-1}u(x)$.

6. Equivariant correspondences and divisibility of local terms

6.1. Notation

We return to the notation of Part I and also fix a finite group G. We are given S-schemes with a left action of G, denoted $\underline{X}$ and $\underline{Y}$, together with stable closed subschemes

$$\underline{A}, \underline{B} \subset \underline{X} \times_S \underline{Y}.$$

We write X, Y, A, B for the underlying schemes, equivalently for the quotients by G in the sense used here. We assume that A and B satisfy the general-position hypotheses of I, 1.2. We are also given objects

$$L_X \in \operatorname{ob} D(X), \qquad L_Y \in \operatorname{ob} D(Y),$$

carrying actions of G above the actions on X and Y, and G-equivariant homomorphisms

$$u : a_1^* L_X \longrightarrow a_2^! L_Y, \qquad v : b_2^* L_Y \longrightarrow b_1^! L_X.$$

Then there are local terms

$$\langle u_z, v_z \rangle \in \Lambda$$

as in I, 1.1.1. On the other hand, the global cup-product $\langle \underline{u}, \underline{v} \rangle$ of I, 1.1.2 belongs to

$$H_c^4(X \times_S Y, K_{X \times_S Y}) = \Lambda,$$

and Theorem 1.2 applies to give

$$\langle u_z, v_z \rangle = \langle \underline{u}, \underline{v} \rangle.$$

One may also consider the non-commutative traces of u_z and v_z. More precisely, $(L_X)_x$ and $(L_Y)_y$ are perfect complexes of $\Lambda[G]$-modules, and u_z, v_z are homomorphisms of $\Lambda[G]$-modules. Thus, by 5.6.3, one can form

$$\operatorname{Tr}_{\Lambda[G]}(u_z v_z) \in \Lambda[G]_\Lambda,$$

where $\Lambda[G]_\Lambda$ is the free Λ-module on the set $G/\sim$ of conjugacy classes of G. Similarly the global cup-product gives an element

$$\langle\!\langle \underline{u}, \underline{v} \rangle\!\rangle \in H_c^4(X \times_S Y, \underline{\Lambda})_{\Lambda[G]},$$

where $\underline{\Lambda}$ denotes the constant sheaf with value $\Lambda[G]_\Lambda$.

Theorem 6.2. With the preceding notation, assume moreover that A and B are in general position in the sense of I, 1.2. Then

$$\mathrm{Tr}_{\Lambda[G]}(u_z v_z) = \langle\!\langle \underline{u}, \underline{v} \rangle\!\rangle.$$

In other words, the non-commutative local term coincides with the non-commutative global term.

The proof follows the same pattern as the proof of Theorem 1.2. By the same reductions as in no. 2, one is first reduced to the case where A and B are closed immersions; the matter is then to prove a vanishing analogous to (2.4.1). The key lemma is the following.

Lemma 6.2.1. Under the hypotheses of 6.2, suppose that $(L_X)|_U = 0$ and $(L_Y)|_V = 0$, with the notation of 2.1. Then

$$\mathrm{Tr}_{\Lambda[G]}(u_z v_z) = \mathrm{Tr}_{\Lambda[G]}(\underline{u} \cdot \underline{v})$$

in $\Lambda[G]_\Lambda$.

The proof of 6.2.1 is analogous to that of 2.3. One is reduced to the case in which L_X and L_Y are concentrated at the closed points, and one applies the non-commutative Lefschetz formula for the projections $\{x\} \to S$ and $\{y\} \to S$.

To finish the proof of 6.2 one still has to establish the non-commutative analogue of 2.5.

Proposition 6.2.2. Under the hypotheses of 6.2, suppose that a and b are closed immersions and that $(L_X)_x = 0$, $(L_Y)_y = 0$. Let T be the strict localization of $X \times_S Y$ at $z = (x, y)$, and identify A and B with closed subschemes of T. Then the restriction map

$$\mathrm{Tr}_{\Lambda[G]} : R\Gamma_A\Big(T, \mathrm{RHom}(DL_X \otimes L_Y, K_T)\Big) \longrightarrow R\Gamma\Big(B, \mathrm{RHom}(DL_X \otimes L_Y, K_T)|_B\Big)$$

is zero in $D(\Lambda[G]_\Lambda)$.

The proof occupies the rest of no. 6. As in no. 3, one is reduced to the tame case, where the representations of inertia factor through ℓ-groups. The reduction to the tame case is analogous to 3.1–3.3. The remaining tame case is treated by means of the purity theorem 4.3 and by an explicit calculation of local terms.

Theorem 6.2 has important consequences for the divisibility of local terms. Let χ be a virtual representation of G over Λ, i.e. an element of $R_\Lambda(G)$. One may consider the local term $\chi(u_z, v_z)$ obtained by applying χ to the non-commutative local term $\mathrm{Tr}_{\Lambda[G]}(u_z v_z)$. More precisely, if $\chi = \sum n_i [V_i]$, with the V_i irreducible representations of G, put

$$\chi(u_z, v_z) = \sum n_i \,\mathrm{Tr}\Big(u_z v_z \mid (L_X)_x \otimes_{\Lambda[G]} V_i\Big).$$

Then 6.2 implies the following.

Corollary 6.3. With the notation of 6.2, let $\chi \in R_\Lambda(G)$, and write

$$e_\chi = \frac{1}{|G|} \sum_{g \in G} \chi(g^{-1}) g$$

for the central idempotent of $\Lambda[G]$ associated with χ. Then

$$\chi(u_z, v_z) = e_\chi \cdot \langle\!\langle \underline{u}, \underline{v} \rangle\!\rangle.$$

In particular, if χ is a representation of degree 1, then

$$\chi(u_z, v_z) = \chi\Big(\langle u_z, v_z \rangle^G\Big),$$

where $\langle u_z, v_z \rangle^G$ denotes the component of $\mathrm{Tr}_{\Lambda[G]}(u_z v_z)$ on the trivial conjugacy class.

When G is cyclic, and one takes for L_X and L_Y the constant sheaves $\Lambda[G]$, and for u and v the correspondences given by left and right multiplication, Theorem 6.2 recovers Grothendieck's divisibility formula (XII, 4.5). More generally, for every finite group G and every virtual representation χ, formula 6.3 gives a canonical way of "dividing" ordinary local terms attached to equivariant correspondences, as announced in the introduction.

Complementary bibliography

[1] H. Cartan and S. Eilenberg, *Homological Algebra*, Princeton University Press, 1956.

[2] A. Grothendieck, "Formule de Lefschetz et rationalité des fonctions L", Séminaire Bourbaki, exposé 279, 1964/65.

[3] J.-L. Verdier, "Catégories dérivées (état 0)", in *SGA* $4\frac{1}{2}$, Lecture Notes in Mathematics 569, Springer, 1977.

[4] R. P. Langlands, "Modular forms and ℓ-adic representations", in *Modular Functions of One Variable II*, Lecture Notes in Mathematics 349, Springer, 1973.

[5] J. Stallings, "Centerless groups – an algebraic formulation of Gottlieb's theorem", *Topology* 4 (1965), 129–134.

[6] M. Artin and J.-L. Verdier, "Refinement of the Thomason–Trobaugh theorem", manuscript.

6.7. Traces and change of scalars

Let X and c be as in 6.5, and let $A \to B$ be a morphism of flat Λ-algebras. With respect to extension and restriction of scalars, the maps Tr_A of 6.5.3, 6.5.8, and 6.5.9 behave like the maps Tr_A of 5.6.1 and 5.6.2. The point is that, if f is an S-morphism, the functors

$$f^*, \quad f_*, \quad f_!, \quad f^!$$

commute with extension and restriction of scalars. Therefore the study of the behavior of Tr_A in 6.5.3 with respect to extension and restriction of scalars is reduced to the behavior of the tensor-product map

$$\mathrm{RHom}_A(p_1^{*L}, p_1^{*KA_X}) \overset{L}{\otimes}_{A\, p_2^{*L}} \longrightarrow \mathrm{RHom}_A(p_1^{*L}, p_1^{*KA_X}) \overset{L}{\otimes}_{A\, p_2^{*L}}$$

and of the evaluation map

$$\mathrm{RHom}_A(L, KA_X) \overset{L}{\otimes}_A L \longrightarrow KA_{X,L_\natural}.$$

Thus the commutativity of the squares 5.9.1 and 5.9.2 implies the commutativity of

$$\begin{array}{ccc} \delta^{*c}{}_*\mathrm{RHom}_A(c_1^{*L}, c_2^{!L}) & \xrightarrow{\mathrm{Tr}_A} & i_*^c KA_{X^c,L_\natural} \\ \downarrow & & \downarrow \\ \delta^{*c}{}_*\mathrm{RHom}_B(c_1^*(B \overset{L}{\otimes}_A L), c_2^!(B \overset{L}{\otimes}_A L)) & \xrightarrow{\mathrm{Tr}_B} & i_*^c KB_{X^c,L_\natural}. \end{array} \tag{6.7.1}$$

Here the left vertical map is extension of scalars and the right vertical map is defined by the canonical morphism

$$A \overset{L}{\otimes}_{A^e} A \longrightarrow B \overset{L}{\otimes}_{B^e} B.$$

It follows that, for $u \in \mathrm{Hom}_A(c_1^{*L}, c_2^{!L})$,

$$\mathrm{Tr}_B(B \overset{L}{\otimes}_A u) = \varphi\, \mathrm{Tr}_A(u), \tag{6.7.2}$$

where

$$\varphi : H^0(X^c, KA_{X^c,L_\natural}) \longrightarrow H^0(X^c, KB_{X^c,L_\natural})$$

is the canonical map.

Assume that A is of finite type and flat as a Λ-module, that B is of finite type and flat as a left A-module, and that $A_\natural$ and $B_\natural$ are flat over Λ. We then have $\mathrm{Tr}_{B/A} : B_\natural \to A_\natural$, and we again write

$$\mathrm{Tr}_{B/A} : KB_{X,\natural} \longrightarrow KA_{X,\natural} \tag{6.7.3}$$

for the map obtained by applying $K_X \overset{L}{\otimes} -$, as in 6.5.2. It follows from 5.10.6 and 5.10.9 that, for $L \in \mathrm{ob}\, D^-_{ctf}(X)$, the square

$$\begin{array}{ccc} \delta^{*c}{}_* \mathrm{RHom}_B(c_1^{*L}, c_2^{!L}) & \xrightarrow{\mathrm{Tr}_B} & i^c_* KB_{X^c,\natural} \\ \downarrow & & \downarrow \mathrm{Tr}_{B/A} \\ \delta^{*c}{}_* \mathrm{RHom}_A(c_1^{*L}, c_2^{!L}) & \xrightarrow{\mathrm{Tr}_A} & i^c_* KA_{X^c,\natural} \end{array} \tag{6.7.4}$$

is commutative, the left vertical map being restriction of scalars. Consequently, for $u \in \mathrm{Hom}_B(c_1^{*L}, c_2^{!L})$, one has in $H^0(X^c, KA_{X^c,\natural})$

$$\mathrm{Tr}_A(u) = \mathrm{Tr}_{B/A}(\mathrm{Tr}_B(u)). \tag{6.7.5}$$

6.8. External products

Let A and B be flat Λ-algebras, let X and Y be S-schemes, and put $Z = X \times_S Y$. Let

$$c : C \to X^2, \qquad d : D \to Y^2$$

be correspondences, and let

$$e = c \times_S d : E = C \times_S D \to Z^2$$

be their product. Then $Z^e = X^c \times_S Y^d$. Put $R = A \otimes_\Lambda B$. As in III, 5.3, one defines the external tensor product

$$\mathrm{Hom}_A(c_1^{*L}, c_2^{!L}) \otimes \mathrm{Hom}_B(d_1^{*M}, d_2^{!M}) \xrightarrow{\overset{L}{\otimes}_S} \mathrm{Hom}_R(e_1^*(L \overset{L}{\otimes}_S M), e_2^!(L \overset{L}{\otimes}_S M)). \tag{6.8.1}$$

From the commutativity of 5.12.1 and 5.12.3 it follows easily that, for $u \in \mathrm{Hom}_A(c_1^{*L}, c_2^{!L})$ and $v \in \mathrm{Hom}_B(d_1^{*M}, d_2^{!M})$, one has in $H^0(Z^e, KR_{Z^e,L_\natural})$

$$\mathrm{Tr}_R(u \overset{L}{\otimes}_S v) = \mathrm{Tr}_A(u)\ \mathrm{Tr}_B(v), \tag{6.8.2}$$

where the product on the right is defined by the canonical map

$$H^0(X^c, KA_{X^c,L_\natural}) \otimes H^0(Y^d, KB_{Y^d,L_\natural}) \longrightarrow H^0(Z^e, KR_{Z^e,L_\natural}). \tag{6.8.3}$$

6.9. G-schemes and equivariant cohomological correspondences

Let G be a finite group. If X is a G-S-scheme, that is, an S-scheme endowed with an action of G by S-automorphisms, and if A is a Λ-algebra, we write $D(G, A^X)$ for the derived category of sheaves of left G-A-modules on X for the étale topology. For the notions of G-sheaves of sets, rings, modules, etc., we refer to XV, 1. We use the subscripts c and ctf for the full subcategories consisting of complexes which, as complexes of A-modules, have constructible cohomology, respectively finite Tor-dimension and constructible cohomology.

Similarly, if B is a Λ-algebra, we write $D(G, A_B^X)$ for the derived category of G-(A, B)-bimodules on X. If G acts trivially on X, then a left G-A-module on X is just a left $A[G]$-module on X without operators. Thus $D(G, A^X) = D(A[G]^X)$. Since G is finite one also has $D_c(A^X) = D_c(A[G]^X)$, but only an inclusion

$$D_{ctf}(A[G]^X) \subset D_{ctf}(G, A^X).$$

The duality formalism of SGA 4, XVII–XVIII extends without difficulty to G-S-schemes. Since the functor forgetting the G-action is faithful, it is enough to extend naturally the definitions of the six operations

$$\overset{L}{\otimes}, \quad \mathrm{RHom}, \quad f^*, \quad f_*, \quad f_!, \quad f^!$$

and of the canonical morphisms considered there (base change, Künneth, duality, etc.). This extension causes no difficulty; for the definition of $f_!$ and $f^!$ one uses equivariant compactifications. We leave to the reader the paraphrase of the definitions and results of 6.2–6.8 in the setting of G-S-schemes.

If, for $i = 1, 2$, X_i is a G-S-scheme, if $c : C \to X = X_1 \times_S X_2$ is a G-S-morphism, and if $L_i \in \operatorname{ob} D_{ctf}(G, A_{X_i}^{X_i})$, the elements of

$$\mathrm{Hom}_A(c_1^{*L_1}, c_2^{!L_2})$$

(where Hom_A means Hom in the relevant equivariant derived category) will be called *equivariant cohomological correspondences from L_1 to L_2 with support in c*. Notice that, when G acts trivially on the X_i and $L_i \in \operatorname{ob} D_{ctf}(A[G]^{X_i})$, the isomorphism 6.4.1 in $D(G, X)$ refines to an isomorphism

$$D_{A[G]} L_1 \overset{L}{\otimes}_{S,A[G]} L_2 \xrightarrow{\sim} \mathrm{RHom}_{A[G]}(p_1^{*L_1}, p_2^{!L_2}).$$

Consequently, in an equivariant situation of type 6.5, with G acting trivially on the spaces and with $L \in \operatorname{ob} D_{ctf}(A[G]^X) \subset D_{ctf}(G, A^X)$, one can, for $u \in \mathrm{Hom}_{A[G]}(c_1^{*L}, c_2^{!L})$, define a local term

$$\mathrm{Tr}_{A[G]}(u) \in H^0(X^c, KA[G]_{X^c, L_\natural}),$$

finer in the evident sense than the local term

$$\mathrm{Tr}_A(u) \in H^0(G, X^c, KA_{X^c, L_\natural})$$

obtained by using only the finite Tor-dimension of L relative to A.

6.10. Values on conjugacy classes

Let X and c be as in 6.5, and let $L \in \mathrm{ob}\, D_{ctf}(A[G]^X)$. For $u \in \mathrm{Hom}_{A[G]}(c_1^{*L}, c_2^{!L})$, 6.5.9 gives a local trace

$$\mathrm{Tr}_{A[G]}(u) \in H^0(X^c, KA[G]_{X^c,\natural}) = H^0(X^c, K_{X^c} \otimes A[G_\natural]), \tag{6.10.1}$$

where $G_\natural$ is the set of conjugacy classes of G. For every $x \in G_\natural$ one may consider the value $\mathrm{Tr}_{A[G]}(u)(x) \in H^0(X^c, K_{X^c})$, obtained by projecting onto the component indexed by x. From 6.7.5 and 5.11.2 one obtains

$$\mathrm{Tr}_A(u) = |G| \ \mathrm{Tr}_{A[G]}(u)(e), \tag{6.10.2}$$

where e denotes the identity element of G.

More generally, let $g \in G$, let Z_g be the centralizer of g, and let $gu \in \mathrm{Hom}_{A[Z_g]}(c_1^{*L}, c_2^{!L})$ be the correspondence obtained by composing u with left multiplication by g on c_1^{*L}. Then $L \in \mathrm{ob}\, D_{ctf}(A[Z_g]^X)$, and 6.10.2 gives

$$\mathrm{Tr}_A(gu) = |Z_g| \ \mathrm{Tr}_{A[Z_g]}(gu)(e). \tag{6.10.3}$$

By the same argument as in 5.11.3, 6.7.4 gives

$$\mathrm{Tr}_{A[Z_g]}(gu)(e) = \mathrm{Tr}_{A[G]}(u)(g^{-1}), \tag{6.10.4}$$

and hence, if desired,

$$\mathrm{Tr}_A(gu) = |Z_g| \ \mathrm{Tr}_{A[G]}(u)(g^{-1}). \tag{6.10.5}$$

6.11–6.13. Direct images and ℓ-adic passage to the limit

Let $f : X \to Y$ be a proper G-S-morphism, and let

$$\begin{array}{ccc} X \times_S X & \xleftarrow{c} & C \\ \downarrow_{f \times_S f} & & \downarrow \\ Y \times_S Y & \xleftarrow{d} & D \end{array} \qquad (6.11.1)$$

be a commutative equivariant square of G-S-schemes, with c and d proper. Assume that G acts trivially on Y and on D. Write

$$f^{c/d}, \text{ or } f^c : X^c \longrightarrow Y^d, \qquad (6.11.2)$$

for the morphism induced on schemes of fixed points. Let $L \in \operatorname{ob} D_{ctf}(G, X)$ and let $u \in \operatorname{Hom}(c_1^{*L}, c_2^{!L})$ be an equivariant correspondence with support in c. Its direct image is an equivariant correspondence

$$f_* u \in \operatorname{Hom}_{A[G]}(d_1^* f_* L, d_2^! f_* L).$$

For $g \in G$, write Z_g for the centralizer of g, and let

$$gu \in \operatorname{Hom}((gc)_1^{*L}, c_2^{!L}) \qquad (6.11.3)$$

be the Z_g-equivariant correspondence with support in

$$gc = (gc_1 = c_1 g, c_2) : C \longrightarrow X \times_S X \qquad (6.11.4)$$

defined as the composite

$$g^*(c_1^{*L}) \xrightarrow{\alpha_g} c_1^{*L} \xrightarrow{u} c_2^{!L},$$

where α_g is the map giving the action of G on c_1^{*L}. It is clear that

$$f_*(gu) = g(f_* u). \qquad (6.11.5)$$

Proposition 6.12. Under the hypotheses of 6.11, assume that $f_* L \in \operatorname{ob} D_{ctf}(A[G]^Y)$. Then, for every $g \in G$,

$$f_*^{gc} \operatorname{Tr}_A(gu) = |Z_g| \operatorname{Tr}_{A[G]}(f_* u)(g^{-1}), \qquad (6.12.1)$$

where

$$f_*^{gc} : H^0(X^{gc}, K_{X^{gc}}) \longrightarrow H^0(Y^d, K_{Y^d})$$

is the Gysin morphism defined by the adjunction map $f_*^{gc} K_{X^{gc}} = f_*^{gc} f^{gc!} K_{Y^d} \to K_{Y^d}$.

Indeed, applying 6.10.5 to f_*u and using 6.11.5 gives

$$\mathrm{Tr}_A(f_*(gu)) = |Z_g|\ \mathrm{Tr}_{A[G]}(f_*u)(g^{-1}),$$

and 6.12.1 follows from the Lefschetz formula III, 4.5.1.

Let now $\ell \neq p$ be a prime. Put $\Lambda_n = \mathbb{Z}/\ell^n\mathbb{Z}$ for $n \geq 1$. Under the hypotheses of 6.11, let $L = (L_n)$ be a projective system of objects of $D_{ctf}(G, X, \Lambda_n)$ such that

$$L_m \overset{L}{\otimes}_{\Lambda_m} \Lambda_n \xrightarrow{\sim} L_n \qquad (m \geq n),$$

and let $u = (u_n)$ be a projective system of equivariant correspondences $u_n \in \mathrm{Hom}(c_1^{*L_n}, c_2^{!L_n})$, compatible under extension of scalars. By compatibility of local traces with extension of scalars, the local terms

$$\mathrm{Tr}_{\Lambda_n}(gu_n) \in H^0(X^{gc}, K\Lambda_{n,X^{gc}})$$

define an element

$$\mathrm{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(X^{gc}, K_{\ell,X^{gc}}) := \varprojlim_n H^0(X^{gc}, K\Lambda_{n,X^{gc}}),$$

and the terms $f_*^{gc}\,\mathrm{Tr}_{\Lambda_n}(gu_n) = \mathrm{Tr}_{\Lambda_n}(gf_*u_n)$ define an element

$$f_*\,\mathrm{Tr}_{\mathbb{Z}_\ell}(gu) \in H^0(Y^d, K_{\ell,Y^d}) := \varprojlim_n H^0(Y^d, K\Lambda_{n,Y^d}).$$

Assume moreover that $f_*M_n \in \mathrm{ob}\, D_{ctf}(\Lambda_n[G]^Y)$ for all n. Then the local traces $\mathrm{Tr}_{\Lambda_n[G]}(f_*u_n)$ likewise define

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_*u) \in H^0(Y^d, K\mathbb{Z}_\ell[G]_{Y^d,\natural}) := \varprojlim_n H^0(Y^d, K\Lambda_n[G]_{Y^d,\natural}).$$

It follows from 6.12 that

$$f_*^{gc}\,\mathrm{Tr}_{\mathbb{Z}_\ell}(gu) = |Z_g|\ \mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_*u)(g^{-1}). \tag{6.13.1}$$

6.14–6.17. Open immersions and divisibility

Let X be a G-S-scheme, let $j : U \to X$ be an equivariant open immersion, and let $c : C \to X \times_S X$ be a morphism of G-S-schemes such that $c_2 = p_2c : C \to X$ is quasi-finite and of finite Tor-dimension. Assume that $c_1^{-1}(U) \subset c_2^{-1}(U)$. The trace morphism

$$\mathrm{Tr}_{c_2} : c_{2!}\Lambda_C \longrightarrow \Lambda_X$$

defines an equivariant filtered correspondence

$$\underline{c} \in \mathrm{Hom}(c_1^* \Lambda_X, c_2^! \Lambda_X), \tag{6.14.2}$$

where Λ_X is endowed with the filtration defined by the submodule $i_! \Lambda_U$, namely $F^n \Lambda_X = \Lambda_X$ for $n \leq 0$, $F^1 \Lambda_X = i_! \Lambda_U$, and $F^n \Lambda_X = 0$ for $n \geq 1$. Its associated graded object consists of two equivariant correspondences

$$\mathrm{gr}^0 \underline{c} = \underline{c}_{X-U,X} \in \mathrm{Hom}(c_1^* \Lambda_{X-U,X}, c_2^! \Lambda_{X-U,X}), \ \mathrm{gr}^1 \underline{c} \quad = \underline{c}_{U,X} \in \mathrm{Hom}(c_1^* \Lambda_{U,X}, c_2^! \Lambda_{X-U,X}), \tag{6.14.3}$$

where $\Lambda_{X-U,X} = j_* \Lambda_{X-U}$ and $\Lambda_{U,X} = i_! \Lambda_U$. Moreover

$$\underline{c}_{X-U,U} = j_* \underline{c}_{X-U}, \tag{6.14.4}$$

where $\underline{c}_{X-U}$ is the correspondence defined by $\mathrm{Tr}_{c_2, X-U}$.

Forgetting the action of G, one has local terms

$$\mathrm{Tr}_\Lambda(\underline{c}), \quad \mathrm{Tr}_\Lambda(\underline{c}_{U,X}), \quad \mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}) \in H^0(X^c, K_{X^c}), \tag{6.14.5}$$

which, by additivity of traces (III, 4.13.1), satisfy

$$\mathrm{Tr}_\Lambda(\underline{c}) = \mathrm{Tr}_\Lambda(\underline{c}_{U,X}) + \mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}). \tag{6.14.6}$$

Furthermore, by 6.14.4,

$$\mathrm{Tr}_\Lambda(\underline{c}_{X-U,X}) = j_*^c \, \mathrm{Tr}_\Lambda(\underline{c}_{X-U}), \tag{6.14.7}$$

where j_*^c is the direct-image morphism attached to $j^c : (X-U)^c \to X^c$.

Recall that if X is smooth, c is a closed immersion, and X^c is finite, then $\mathrm{Tr}_\Lambda(\underline{c})$ is the image in Λ of the intersection multiplicity $(C \cdot X)$ of C with the diagonal. If moreover $X-U$ is smooth, then $\mathrm{Tr}_\Lambda(\underline{c}_{X-U})$ is the image in Λ of $(c_2^{-1}(X-U) \cdot (X-U))$. Also, if $X-U$ is proper over S and $c|_{X-U}$ is the diagonal, the Lefschetz formula III, 4.7 gives that

$$\int_{X-U} \mathrm{Tr}(\underline{c}_{X-U}) \in \Lambda$$

is the Euler-Poincaré characteristic of $X-U$ with values in Λ.

Now let a diagram 6.11.1 be given, with f, c, d proper, and let $U \to X$ be an equivariant open immersion. Assume that c_2 is quasi-finite and of finite Tor-dimension, that $c_1^{-1}(U) \subset c_2^{-1}(U)$, that G acts trivially on C and D, and that

$$f|_U : U \longrightarrow f(U) = V$$

is a Galois étale covering with group G. This last hypothesis implies that $Rf_*(\Lambda_{U,X}) \in \operatorname{ob} D_{ctf}(A[G]^Y)$; indeed it is concentrated in degree zero and is the extension by zero of a sheaf on V locally isomorphic to $\Lambda[G]$. Thus one can apply 6.12 to $L = \Lambda_{U,X}$ and $u = \underline{c}_{U,X}$, and, using 6.14.6 and 6.14.7, one gets:

Proposition 6.16. Under the hypotheses of 6.15, for every $g \in G$,

$$f_*^{gc} \operatorname{Tr}_\Lambda(\underline{gc}) - (fj)_*^{gc} \operatorname{Tr}_\Lambda(\underline{gc}_{X-U}) = |Z_g| \operatorname{Tr}_{A[G]}(f_* \underline{c}_{U,X})(g^{-1}), \tag{6.16.1}$$

where $j : X - U \to X$ is the inclusion and Z_g is the centralizer of g.

Applying 6.16 to the projective system of correspondences $\underline{c}_\ell = (\underline{c}^{\Lambda_n})$, with $\Lambda_n = \mathbb{Z}/\ell^n$ and $\ell \neq p$, and passing to the limit in 6.14.6 and 6.14.7, gives:

Corollary 6.17. Under the hypotheses of 6.15, for every $g \in G$,

$$f_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(\underline{gc}_\ell) - (fj)_*^{gc} \operatorname{Tr}_{\mathbb{Z}_\ell}(\underline{gc}_{\ell,X-U}) = |Z_g| \operatorname{Tr}_{\mathbb{Z}_\ell[G]}(f_* \underline{c}_{\ell,U,X})(g^{-1}), \tag{6.17.1}$$

with the notation of 6.16.

Taking into account compatibility of local traces with localization and the remarks made in 6.14, one sees that 6.17 gives an affirmative answer, in a more general formulation, to Grothendieck's divisibility conjecture (XII, 4.5) in the case where the endomorphism φ of G considered there is the identity.

In fact, inspired by 5.13, it is not difficult to formulate analogous divisibility statements to 6.12, 6.13, 6.16, and 6.17 in the case where c is not G-equivariant but c_2 is, and c_1 satisfies

$$c_1(xg) = c_1(x)\varphi(g)$$

for every point x of G, with a fixed endomorphism φ of G. The formulae then read as above, with Z_g replaced by the φ-centralizer of g^{-1} (5.13.5).

6.18–6.24. Invariants and the comparison with Grothendieck local terms

Put

$$N_G = \sum_{g \in G} g \in \Lambda[G]. \tag{6.18.1}$$

Let X be an S-scheme. For $L \in \mathrm{ob}\, D(A[G]^X)$, left multiplication by N_G induces a morphism in $D(X)$

$$N_G : \Lambda \overset{L}{\otimes}_{A[G]} L \longrightarrow \mathrm{RHom}_{A[G]}(\Lambda, L), \tag{6.18.2}$$

where Λ is regarded as a $\Lambda[G]$-algebra through the natural augmentation sending every g to 1.

Lemma 6.18.3. If $L \in \mathrm{ob}\, D_{ctf}(A[G]^X)$, the morphism 6.18.2 is an isomorphism.

By dévissage one is reduced to the case in which L is perfect, then to $L = \Lambda[G]$. The assertion then follows from the fact that N_G is a basis of the invariant module of the left $\Lambda[G]$-module $\Lambda[G]$.

Proposition 6.19. In the situation of 6.11, let $M \in \mathrm{ob}\, D_{ctf}(Y)$, let $v \in \mathrm{Hom}(d_1^{*M}, d_2^{!M})$, and let $a : M \to f_*L$ be a morphism in $D(A[G]^Y)$, where M is regarded as an object of $D(A[G]^Y)$ through the augmentation $\Lambda[G] \to \Lambda$. Assume that the square

$$\begin{array}{ccc} d_1^{*M} & \overset{v}{\to} & d_2^{!M} \\ \downarrow d_1^{*a} & & \downarrow d_2^{!a} \\ d_1^* f_* L & \overset{f_*u}{\longrightarrow} & d_2^! f_* L \end{array}$$

is commutative. Assume moreover that $f_*M \in \mathrm{ob}\, D_{ctf}(A[G]^Y)$ and that a induces an isomorphism

$$M \xrightarrow{\sim} \mathrm{RHom}_{A[G]}(\Lambda, f_*L).$$

Then

$$\mathrm{Tr}_\Lambda(v) = \sum_{g \in G_\natural} \mathrm{Tr}_{A[G]}(f_*u)(g^{-1}). \tag{6.19.1}$$

Indeed, by 6.18.3, composing a with N_G^{-1} gives an isomorphism

$$M \xrightarrow{\sim} \Lambda \overset{L}{\otimes}_{A[G]} f_*L$$

under which v is identified with $\Lambda \overset{L}{\otimes}_{A[G]} f_*u$. By 6.7.2, $\mathrm{Tr}_\Lambda(v)$ is the image of $\mathrm{Tr}_{A[G]}(f_*u)$ under the map induced by the augmentation $\Lambda[G] \to \Lambda$. This map sends each conjugacy class to 1, which proves the formula.

Let X_i for $i = 1, 2$ be G-S-schemes, put $X = X_1 \times_S X_2$, and let $c : C \to X$ be a G-S-morphism such that c_2 is quasi-finite and of finite Tor-dimension. For $L_i \in \mathrm{ob}\, D^+(G, X_i)$, the trace morphism $\mathrm{Tr}_{c_2} : c_{2!} c_2^{*L_2} \to L_2$ defines by adjunction a map, again denoted

$$\mathrm{Tr}_{c_2} : c_2^{*L_2} \longrightarrow c_2^{!\,L_2}, \tag{6.20.1}$$

and hence every equivariant morphism $u \in \mathrm{Hom}(c_1^{*L_1}, c_2^{*L_2})$ determines an equivariant correspondence

$$u^! = \mathrm{Tr}_{c_2} \circ u \in \mathrm{Hom}(c_1^{*L_1}, c_2^{!\,L_2}). \tag{6.20.2}$$

Proposition 6.21. Under the hypotheses of 6.11, assume moreover that an open immersion $i : V \hookrightarrow Y$ is given such that $(fc_1)^{-1}(V)$ is connected, that $d_1^{-1}(V) \subset d_2^{-1}(V)$, and that f induces a principal Galois covering with group G,

$$U = f^{-1}(V) \xrightarrow{f_V} V.$$

Assume also that $M = i_!E$ with $E \in \mathrm{ob}\, D(V)$, and that one has an isomorphism

$$\alpha : f_V^* E \xrightarrow{\sim} \Lambda_X \otimes_\Lambda E_0$$

in $D(G, X)$, with $E_0 \in \mathrm{ob}\, D(\Lambda[G])$ perfect over Λ. Let $v \in \mathrm{Hom}(d_1^{*M}, d_2^{*M})$ be as above, and let $v_0 \in \mathrm{Hom}_{\Lambda[G]}(E_0, E_0)$ be the endomorphism defined, via α, by $f^*v|_{c_1^{-1}(U)}$. If $\underline{c}_{U,X}$ denotes the correspondence of 6.14.3, then, with the notation 6.20.2,

$$\mathrm{Tr}_\Lambda(v^!) = \sum_{g \in G_\natural} \mathrm{Tr}_{A[G]}(f_*\underline{c}_{U,X})(g^{-1})\ \mathrm{Tr}_\Lambda(gv_0). \tag{6.23.1}$$

Because of the central character of Tr_Λ, the right hand side does not depend on the choice of α. The projection formula $f_*f^*\Lambda_Y \otimes M \xrightarrow{\sim} f_*f^*M$ and 6.15 show that M satisfies the hypotheses needed above. The isomorphism α identifies $(f^*v)^!$ with the external tensor product $\underline{c}_{U,X} \otimes v_0$. Thus f_*f^*M is identified with $f_*(\Lambda_{U,X}) \otimes E_0$, and $f_*(f^*v)^!$ with $f_*\underline{c}_{U,X} \otimes v_0$. For $g \in G$, the Z_g-equivariant correspondence $gf_*(f^*v)^!$ is identified with $gf_*\underline{c}_{U,X} \otimes gv_0$. Since 6.10.4 gives

$$\mathrm{Tr}_{A[G]}(f_*(f^*v)^!)(g^{-1}) = \mathrm{Tr}_{A[Z_g]}(gf_*(f^*v)^!)(e),$$

it is enough, using 6.22.1, to prove

$$\mathrm{Tr}_{A[Z_g]}(gf_*\underline{c}_{U,X} \otimes gv_0)(e) = \mathrm{Tr}_{A[G]}(f_*\underline{c}_{U,X})(g^{-1})\ \mathrm{Tr}_\Lambda(gv_0),$$

or equivalently, by 6.10.4,

$$\mathrm{Tr}_{A[Z_g]}(gf_*\underline{c}_{U,X} \otimes gv_0)(e) = \mathrm{Tr}_{A[Z_g]}(gf_*\underline{c}_{U,X})(e)\ \mathrm{Tr}_\Lambda(gv_0). \tag{6.23.2}$$

Replacing G by Z_g, one may suppose $g = e$, and then, returning to the definition of traces, it suffices to establish the following lemma.

Lemma 6.23.3. Let P be a G_V-module locally a direct factor of a free module of finite type, put $P' = i_! P$, and let Q be a complex of $\Lambda[G]$-modules perfect over Λ. Then $P \overset{L}{\otimes}_\Lambda Q$, with the diagonal action of G, is perfect over $\Lambda[G]$, and the square

$$\begin{array}{ccc} \delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^{*P'}, p_2^{!P'}) \overset{L}{\otimes} \mathrm{RHom}_{\Lambda[G]}(Q, Q) & \xrightarrow{(1)} & K_Y \\ \downarrow{\scriptstyle (*)} & & \| \\ \delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^{*P'} \overset{L}{\otimes} Q, p_2^{!P'} \overset{L}{\otimes} Q) & \xrightarrow{(3)} & K_Y \end{array}$$

is commutative. Here $\delta : Y \to Y \times_S Y$ is the diagonal; (1) is the tensor product of

$$\delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^{*P'}, p_2^{!P'}) \xrightarrow{6.5.3} K\Lambda[G]_{Y,\natural} \xrightarrow{x \mapsto x(e)} K_Y$$

and

$$\mathrm{RHom}_{\Lambda[G]}(Q, Q) \xrightarrow{\mathrm{Tr}_\Lambda} \Lambda;$$

$(*)$ is induced from the external tensor product by restriction along the diagonal $\Lambda[G] \to \Lambda[G] \otimes \Lambda[G]$; and (3) is 6.5.3 followed by $x \mapsto x(e)$.

The first assertion is clear: one is reduced to $P = \Lambda[G]_V$, and then $P \overset{L}{\otimes}_\Lambda Q$ is the tensor product of P by Q with trivial G-action. For the second assertion, 6.5.4 and 6.5.5 give

$$\delta^* \mathrm{RHom}_{\Lambda[G]}(p_1^{*P'}, p_2^{!P'}) \xrightarrow{\sim} D_{\Lambda[G]}(i_! P) \overset{L}{\otimes}_{\Lambda[G]} i_! P \xrightarrow{\sim} i_!(D_{\Lambda[G]} P \overset{L}{\otimes}_{\Lambda[G]} P),$$

so one is reduced to checking the induced square on V; thus one may assume $V = Y$. The square becomes

$$\begin{array}{ccc} \mathrm{RHom}_{\Lambda[G]}(P, K_Y \overset{L}{\otimes} P) \overset{L}{\otimes} \mathrm{RHom}_{\Lambda[G]}(Q, Q) & \xrightarrow{(1)} & K_Y \\ \downarrow & & \| \\ \mathrm{RHom}_{\Lambda[G]}(P \overset{L}{\otimes} Q, K_Y \overset{L}{\otimes} P \overset{L}{\otimes} Q) & \xrightarrow{(3)} & K_Y. \end{array}$$

Here (1) is given by $\mathrm{Tr}_{\Lambda[G]}(-)(e) \otimes \mathrm{Tr}_\Lambda$, and (3) by $\mathrm{Tr}_{\Lambda[G]}(-)(e)$. By descent one is reduced to $P = \Lambda[G]_Y$, and then one may suppose that G acts trivially on Q. The assertion is then an immediate consequence of 5.12.5.

If Λ is killed by a power of a prime $\ell \neq p$, the right hand side of 6.23.1 may be rewritten as

$$\sum_{g \in G_\natural} \mathrm{Tr}_{\mathbb{Z}_\ell[G]}(f_* \underline{c}_{\ell,U,X})(g^{-1}) \; \mathrm{Tr}_\Lambda(g v_0),$$

with the notation of 6.17. Formula 6.17.1 shows that Grothendieck's local terms in XII, 6.2 coincide with Verdier's local terms, at least when the endomorphism φ of G considered there is the identity; as already observed, this restriction is not serious. If Y is proper over S, combining 6.23.1 with the Lefschetz formula for $Y \to S$ (III, 4.7) gives a formula generalizing XII, 6.3, and hence in particular the Lefschetz formula for Frobenius proved in SGA $4\frac{1}{2}$, Rapport, and in Part I of the present exposé.

Bibliography

[1] H. Cartan and S. Eilenberg, *Homological Algebra*, Princeton University Press, 1956.

[2] A. Grothendieck, *Formule de Lefschetz et rationalité des fonctions L*, Séminaire Bourbaki 1964/65, no. 279, in *Dix exposés sur la cohomologie des schémas*, North Holland, 1968.

[3] L. Illusie, *Complexe cotangent et déformations I, II*, Lecture Notes in Mathematics 239 and 283, Springer, 1971–1972.

[4] R. P. Langlands, *Modular forms and ℓ-adic representations*, in *Modular Functions of One Variable II*, Lecture Notes in Mathematics 349, Springer, 1973.

[5] J. Stallings, *Centerless groups – an algebraic formulation of Gottlieb's theorem*, Topology 4 (1965), 129–134.

[6] J.-L. Verdier, *The Lefschetz fixed point formula in étale cohomology*, in *Proceedings of a Conference on Local Fields*, Springer, 1967.

SGA $4\frac{1}{2}$: *Cohomologie étale*, by P. Deligne, Lecture Notes in Mathematics 569, Springer, 1977. In SGA $4\frac{1}{2}$: *Th. Finitude* refers to the finiteness theorems in ℓ-adic cohomology; *Cycle* to the cohomology class associated with a cycle; and *Rapport* to the report on the trace formula.

Exposé V. J-adic projective systems

J. P. Jouanolou

1. Generalities on abelian A-categories

1.1. Definitions and first properties

Let A be a commutative ring. An *A-category* $\mathcal{C}$ is a category for which, for each pair (X, Y) of objects, the set $\mathrm{Hom}(X, Y)$ is endowed with an A-module structure, compatibly with composition. Thus for three objects X, Y, Z the composition map

$$\mathrm{Hom}(Y, Z) \times \mathrm{Hom}(X, Y) \longrightarrow \mathrm{Hom}(X, Z)$$

is A-bilinear.

An *abelian A-category* is an additive A-category in which every morphism has a kernel and a cokernel and in which every morphism factors canonically as an epimorphism followed by a monomorphism. More precisely, one assumes an equivalence between $\mathcal{C}$ and a full subcategory of the category of sheaves of A-modules on a site, so that finite inverse and direct limits, kernels, cokernels, images, and coimages are representable and are computed sheafwise.

An *A-functor*, respectively an additive A-functor, between A-categories is a functor, respectively an additive functor, whose action on Hom-sets is A-linear. An *(A, B)-functor* from an A-category to a B-category consists of a ring homomorphism $A \to B$ and a B-functor to the target. When $A = B$ and the map is the identity, this is simply an A-functor.

Let $\mathcal{C}$ be an abelian A-category. An object M of $\mathcal{C}$ is said to be of *finite type* (respectively *noetherian*) if, for every filtered inductive system $(N_i)_{i \in I}$ of subobjects of M, the canonical morphism

$$\varinjlim_i N_i \longrightarrow M$$

has the expected finiteness behavior: in the finite-type case the union stabilizes as soon as it is all of M, and in the noetherian case every increasing sequence of subobjects is stationary. An object is *coherent* if it is of finite type and if, for every map $u : N \to M$ with N of finite type, $\mathrm{Ker}(u)$ is of finite type. For ordinary modules these are the usual notions.

If J is an ideal of A, the *canonical J-filtration* of an object M is the decreasing filtration by the images $J^n M$ of $J^n \otimes_A M \to M$. An object is called *J-adic* if this filtration is J-adic: for every n, the canonical map $J \otimes_A J^n M \to J^{n+1} M$ is an epimorphism and the transition maps $M/J^{n+1}M \to M/J^nM$ are epimorphisms.

Throughout the exposé, the ideal J is assumed to satisfy the Artin–Rees condition (AR) of 1.1.8(ii). We shall use without further comment the elementary facts that the full subcategory of noetherian objects is thick and abelian: it is stable under subobjects, quotients, and extensions.

1.2–1.3. Projective systems and AR-zero systems

Let $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ denote the category of projective systems indexed by $\mathbb{N}$ with values in $\mathcal{C}$. A system $X = (X_n)$ is called *AR-zero* if for every n there is an integer $m \geq n$ such that the transition map $X_m \to X_n$ is zero. The AR-zero systems form a thick subcategory, and in fact a localizing subcategory, of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ in the sense of Gabriel.

The quotient category

$$\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}) / \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})_{\text{AR-zero}}$$

is therefore abelian, and the canonical functor is exact. If $\mathcal{C}'$ is a full thick abelian subcategory of $\mathcal{C}$, one writes $\mathrm{Hom}_{\mathrm{AR}}(\mathbb{N}^\circ, \mathcal{C}')$ for the full subcategory of systems which are AR-isomorphic to systems with values in $\mathcal{C}'$. The usual properties of quotient categories then give the corresponding thick abelian subcategory in the quotient.

1.4. Examples and constructions

For an object M of $\mathcal{C}$, let $\kappa(M)$ denote the constant projective system with value M. The functor

$$\kappa : \mathcal{C} \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$$

is fully faithful and exact.

For an integer $r \geq 0$, the translation functor $[r]$ on projective systems is defined by

$$(X[r])_n = X_{n+r}.$$

It is exact, and there is a canonical monomorphism $X[r] \to X$ whose cokernel is AR-zero.

If $\mathcal{C}$ is complete, the projective limit

$$\varprojlim X = \varprojlim_n X_n$$

is an exact-left functor. For ordinary countable systems one has the standard exact sequence involving $R^1 \varprojlim$, and the higher derived functors vanish under the hypotheses used below.

The separated J-adic completion of an object M is

$$\widehat{M} = \varprojlim_{n} M/J^n M$$

when it exists. If $\mathcal{C}$ is complete and countable products are exact, completion is exact on noetherian objects. A projective system X is *strict J-adic* if X_n is killed by J^{n+1}, the transition maps are epimorphisms, and $X_m/J^{n+1}X_m \xrightarrow{\sim} X_n$ for $m \geq n$. Such a system is noetherian, respectively artinian, if its components are noetherian, respectively artinian.

The following elementary lemmas will be used repeatedly. A noetherian object has all subobjects noetherian. If M is separated and complete for the J-adic topology, then the system $(M/J^n M)_n$ is strict J-adic, and its image in the quotient by AR-zero systems is isomorphic to the image of the constant system $\kappa(M)$.

2. Mittag-Leffler–Artin–Rees and calculus of fractions

In this section $\mathcal{C}$ is an abelian A-category and $J \subset A$ satisfies (AR). A projective system $X = (X_n)$ satisfies the *Mittag-Leffler–Artin–Rees condition*, abbreviated (MLAR), if for every n there is $m \geq n$ such that for all $p \geq m$ the image of $X_p \to X_n$ is equal to the image of $X_m \to X_n$. In other words, the family of images

$$\mathrm{Im}(X_m \to X_n)_{m \geq n}$$

is stationary. A stricter version asks in addition that this stable image be a J-adic object of $\mathcal{C}$.

A system satisfying (MLAR) is AR-isomorphic to a strict projective system whose transition maps are epimorphisms. The condition is stable under short exact sequences in the familiar way. A J-adic projective system satisfies (MLAR) if and only if it is strict. Every noetherian projective system satisfies (MLAR).

Let X be a strict J-adic system. Its kernel is

$$\mathrm{Ker}(X) = (\mathrm{Ker}(X_n \to X_0))_n.$$

This is a strict subsystem, and the quotient is the constant system $\kappa(X_0)$. If Y is a strict subsystem of X, we say that Y is *J-adapted* to X if for every n,

$$Y_n = X_n \cap Y_0 \quad \text{inside } X_0.$$

For such a subsystem, Y is J-adic exactly when the associated graded system is a module over $\mathrm{gr}_J(A)$ in the evident sense.

The associated graded object of a strict J-adic system is

$$\mathrm{gr}(X) = \bigoplus_{n \geq 0} X_n / X_{n+1},$$

viewed as a graded module over $\mathrm{gr}_J(A)$. The system X is noetherian, respectively artinian, if and only if $\mathrm{gr}(X)$ is noetherian, respectively artinian, as a graded $\mathrm{gr}_J(A)$-module.

Localizing subcategories in the sense of Gabriel will be used to express the calculus of fractions. If $\mathcal{L}$ is a localizing subcategory of an abelian category $\mathcal{A}$, a morphism is a $\mathcal{L}$-equivalence if its kernel and cokernel belong to $\mathcal{L}$. The quotient $\mathcal{A}/\mathcal{L}$ is the localization with respect to these equivalences, and the canonical functor $\mathcal{A} \to \mathcal{A}/\mathcal{L}$ is exact and universal for exact functors annihilating $\mathcal{L}$.

Applied to the localizing subcategory of AR-zero systems, this gives the category in which AR-isomorphisms are inverted. In particular, for a strict J-adic system X, the functor sending a J-adapted strict subsystem $Y \subset X$ to its image in the quotient is an equivalence from the category of such subsystems to the category of subobjects of the image of X in the quotient.

3. J-adic and AR-J-adic projective systems

A projective system is called *J-adic* if it is isomorphic to a strict J-adic system. A system is called *AR-J-adic* if it is AR-isomorphic to a strict J-adic system. We write

$$J\text{-adn}(\mathcal{C}), \quad J\text{-adf}(\mathcal{C}), \quad \text{AR-}J\text{-adn}(\mathcal{C}), \quad \text{AR-}J\text{-adf}(\mathcal{C})$$

for the corresponding noetherian and artinian subcategories.

A J-adic system remains J-adic after any translation $[r]$. Moreover, every AR-J-adic noetherian system is represented, in the quotient by AR-zero systems, by a translate $Y[r]$ of an actual J-adic noetherian system Y.

Assume now that $\mathcal{C}$ is complete and that countable products are exact. For any projective system X, the functor $\varprojlim$ is left exact, and its derived functors satisfy

$$R^q \varprojlim X = 0 \quad (q \geq 2).$$

If X is AR-zero, then $R^q \varprojlim X = 0$ for all $q \geq 0$. If X is AR-J-adic noetherian, then

$$R^1 \varprojlim X = 0.$$

Consequently $\varprojlim X$ is separated and complete for the J-adic topology, and the canonical morphisms

$$X_n \longrightarrow \varprojlim X/J^{n+1} \varprojlim X$$

are isomorphisms.

The J-adic completion functor

$$\Lambda_J(M) = \varprojlim_n M/J^n M$$

is additive and left exact. Its left derived functors satisfy

$$L_i \Lambda_J(M) = 0 \qquad (i \geq 2).$$

For M noetherian, one has $L_1\Lambda_J(M) = 0$, and $\Lambda_J(M) \cong \varprojlim_n M/J^n M$. If $\mathcal{C}$ is complete with exact countable products and M is noetherian, the canonical map

$$\widehat{A} \otimes_A M \longrightarrow \Lambda_J(M)$$

is an isomorphism; this is the familiar exactness of completion for finite modules in this categorical setting.

4. Filtrations and gradings

Let $X = (X_n)$ be a projective system. A filtration of X is a compatible family of decreasing filtrations $F^\bullet X_n$. It is called *J-good* if, for every n, the filtration on X_n is J-good, i.e. for some n_0,

$$J^p F^{n_0} X_n = F^{n_0+p} X_n \qquad (p \geq 0).$$

The associated graded system is

$$\mathrm{gr}(X) = (\mathrm{gr}(X_n))_n, \qquad \mathrm{gr}(X_n) = \bigoplus_{p \geq 0} F^p X_n / F^{p+1} X_n.$$

If X is strict J-adic and has a J-good filtration, then $\mathrm{gr}(X)$ is a strict $\mathrm{gr}_J(A)$-adic projective system. If moreover X is noetherian, so is $\mathrm{gr}(X)$.

The Artin–Rees lemma takes the following form. Let M be a noetherian object of an abelian A-category and $N \subset M$ a subobject. Give M its canonical J-filtration. Then the induced filtration on N is J-adic: for every n there exists m such that

$$N \cap J^m M \subset J^n N.$$

The proof is by applying (MLAR) to the projective system $(N/(N \cap J^n M))_n$.

For a noetherian object M, the canonical map

$$J^n \otimes_A M \longrightarrow J^n M$$

is an epimorphism. Moreover, the system $(J^n M)_n$ is AR-zero if and only if M is killed by a power of J. If an AR-J-adic noetherian system carries a J-good filtration, its associated graded object is AR-$\mathrm{gr}_J(A)$-adic and noetherian. In a short exact sequence of J-adic noetherian systems with J-good filtrations, the associated sequence of graded systems is exact. Finally, for a noetherian strict J-adic system X, there exists $r \geq 0$ such that, for every strict subsystem $Y \subset X$, the graded object $\mathrm{gr}(Y[r])$ is a graded $\mathrm{gr}_J(A)$-module.

The Nakayama lemma also has the expected form. If M is separated and complete for the J-adic topology and $M/JM = 0$, then $M = 0$. Hence a morphism between separated and complete objects is an epimorphism if and only if it is so modulo J. Similarly, M is of finite type or noetherian if and only if M/JM has the corresponding property over A/J, with the usual finiteness condition on the kernels of $J^n \otimes_A M \to J^n M$ in the noetherian criterion.

5. Noetherian AR-J-adic systems

In this section $\mathcal{C}$ is an abelian A-category, J is an ideal satisfying (AR), and $\mathcal{C}$ has countable direct sums.

A projective system $X = (X_n)$ is *AR-J-adic noetherian* (respectively artinian) if it is AR-J-adic and its components are noetherian (respectively artinian). We write

$$\text{AR-}J\text{-adn}(\mathcal{C}), \qquad \text{AR-}J\text{-adf}(\mathcal{C})$$

for the corresponding full subcategories of the AR quotient. The canonical functors

$$p^n_{\mathrm{AR}} : J\text{-adn}(\mathcal{C}) \to \text{AR-}J\text{-adn}(\mathcal{C}), \qquad p^f_{\mathrm{AR}} : J\text{-adf}(\mathcal{C}) \to \text{AR-}J\text{-adf}(\mathcal{C})$$

are equivalences of categories.

We shall need Hilbert's theorem in the following form. Let A be a commutative ring, $\mathcal{C}$ an abelian A-category, and M an object of finite type, respectively noetherian. For every finitely generated A-algebra $B = \bigoplus_{n \geq 0} B_n$ graded by $\mathbb{N}$, the graded $(B, \mathcal{C})$-module $B \otimes_A M$ is of finite type, respectively noetherian. The proof reduces at once to $B = A[T]$, where submodules of $A[T] \otimes_A M$ correspond to increasing arrays of subobjects of M; the assertion is then precisely the finite-type or noetherian condition on M.

If $\mathcal{C}$ has countable direct sums, the same conclusion holds for an arbitrary finitely generated A-algebra B, not necessarily graded. It follows in particular that if A is noetherian and

$J \subset A$ is an ideal, then, for every noetherian J-adic system X, the graded object $\mathrm{gr}(X)$ is a noetherian $(\mathrm{gr}_J(A), \mathcal{C})$-module. Indeed $\mathrm{gr}_J(A)$ is a finitely generated A-algebra and there is an epimorphism

$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{gr}(X).$$

Let $\mathcal{E}_{\mathcal{C}}$ be the full subcategory of $\underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})$ whose objects are AR-J-adic noetherian systems.

Proposition 5.2.1. The category $\mathcal{E}_{\mathcal{C}}$ is stable under kernels and cokernels in $\underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})$. Thus $\mathcal{E}_{\mathcal{C}}$ is an abelian category and the inclusion

$$\mathcal{E}_{\mathcal{C}} \longrightarrow \underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})$$

is exact.

Let $u : X \to Y$ be a morphism in $\mathcal{E}_{\mathcal{C}}$. One proves successively that $\mathrm{Coker}(u)$, $\mathrm{Im}(u)$, and $\mathrm{Ker}(u)$ are AR-J-adic and noetherian. For the image one uses the following lemma.

Lemma 5.2.1.1. Let

$$0 \to L \to M \to N \to 0$$

be exact in $\underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})$. If M and N are AR-J-adic noetherian, then L is AR-J-adic noetherian.

It is enough to prove the AR-J-adic assertion. One reduces as in the proof of 3.2.4 to the case where M and N are strict, and even J-adic noetherian. Then 3.1.3(ii) shows that L is strict. It is adapted to J, and its graded object, as a graded subobject of $\mathrm{gr}(M)$, is noetherian. The assertion follows from the Artin–Rees lemma 4.2.6. Applying this first to $0 \to \mathrm{Im}(u) \to Y \to \mathrm{Coker}(u) \to 0$ and then to $0 \to \mathrm{Ker}(u) \to X \to \mathrm{Im}(u) \to 0$ gives the proposition.

Proposition 5.2.2. The objects of the abelian category AR-J-$\mathrm{adn}(\mathcal{C})$ are noetherian.

It is enough to show that, for a J-adic noetherian object X, any increasing sequence of subobjects in the AR category is stationary. Such a sequence may be represented by morphisms

$$v_n : Y^n \longrightarrow X$$

which are monomorphisms in the AR category and factor through one another there. The assertion that v_n factors through v_{n+1} means that for some r there is a morphism $f : Y^n[r] \to Y^{n+1}$ with

$$v_n \circ a = v_{n+1} \circ f,$$

where $a : Y^n[r] \to Y^n$ is the canonical strict morphism. Hence the image of v_n is contained in that of v_{n+1}. Replacing Y^n by the image, one is reduced to an increasing sequence of strict subsystems of X. Since $\mathrm{gr}(X)$ is noetherian, the increasing sequence of the corresponding graded systems is stationary, and so is the original sequence.

In summary: $\mathcal{E}_{\mathcal{C}}$ is a thick abelian subcategory of $\underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})$; passage to the quotient by AR-zero systems induces an equivalence

$$\mathcal{E}_{\mathcal{C}}/(\mathcal{E}_{\mathcal{C}} \cap \underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})_{\text{AR-zero}}) \xrightarrow{\sim} \text{AR-}J\text{-adn}(\mathcal{C});$$

and the objects of AR-J-adn$(\mathcal{C})$ are noetherian.

Theorem 5.2.3, stated in this language, says that both J-adn$(\mathcal{C})$ and AR-J-adn$(\mathcal{C})$ are abelian and noetherian categories. For AR-J-adn$(\mathcal{C})$ this is precisely what has just been proved; the category J-adn$(\mathcal{C})$ is equivalent to it.

Proposition 5.2.4. Stability under extensions. Let

$$0 \to X \to Y \to Z \to 0$$

be exact in $\underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for all n. Assume that Z is AR-J-adic and that X is AR-J-adic artinian. Then Y is AR-J-adic.

As in the proof of 3.2.4, one reduces to the case in which Y and Z are strict and Z is J-adic. Then X is strict. For some r the projective system

$$(X_{n+r}/J^{n+1}X_{n+r})_{n\in\mathbb{N}}$$

is J-adic artinian. Since X is a quotient of it and is AR-isomorphic to it, one is reduced to the following lemma.

Lemma 5.2.4.1. Let

$$0 \to T \to X \to Y \to Z \to 0$$

be exact in $\underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for all n. If X is J-adic artinian, Z is J-adic, and T is AR-zero, then Y is AR-J-adic.

For fixed n, the evident system

$$(Y_{n+k}/J^{n+1}Y_{n+k})_{k\in\mathbb{N}}$$

is essentially constant. Indeed, let S_n^k be the kernel of

$$X_{n+k}/J^{n+1}X_{n+k} \longrightarrow Y_{n+k}/J^{n+1}Y_{n+k}.$$

In the exact commutative diagram

$$\begin{array}{ccccccccc}
0 & \to & S_n^k & \to & X_{n+k}/J^{n+1}X_{n+k} & \to & Y_{n+k}/J^{n+1}Y_{n+k} & \to & Z_{n+k}/J^{n+1}Z_{n+k} \to 0 \\
 & & \downarrow & & \downarrow \wr & & \downarrow & & \downarrow \wr \\
0 & \to & T_n & \to & X_n & \to & Y_n & \to & Z_n \to 0,
\end{array}$$

the S_n^k are subobjects of the artinian object T_n and form a decreasing sequence. Hence they are eventually constant, which gives the essential constancy.

Define a system Y' by

$$Y'_n = \varprojlim_r Y_{n+r}/J^{n+1}Y_{n+r}.$$

Then Y' is J-adic. Passing to the limit gives an exact commutative diagram

$$\begin{array}{ccccccccc}
0 & \to & S & \to & X & \to & Y' & \to & Z \to 0 \\
 & & \downarrow & & \downarrow \mathrm{id} & & \downarrow & & \downarrow \mathrm{id} \\
0 & \to & T & \to & X & \to & Y & \to & Z \to 0.
\end{array}$$

Since T is AR-zero, the snake lemma shows that $Y' \to Y$ is an AR-isomorphism. Thus Y is AR-J-adic.

Corollary 5.2.5. Let

$$0 \to X \to Y \to Z \to 0$$

be exact in $\underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for all n. Assume that X is AR-J-adic artinian, respectively artinian and noetherian, and that Z is AR-J-adic artinian, respectively noetherian. Then Y is AR-J-adic artinian, respectively noetherian.

5.3. AR-J-adic noetherian systems and ∂-functors

Let $\mathcal{C}$ and $\mathcal{D}$ be two abelian A-categories and let $F : \mathcal{C} \to \mathcal{D}$ be an additive functor. It extends termwise to an additive functor

$$F : \underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C}) o \underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{D})$$

which commutes with translations.

Proposition 5.3.1. Let $\mathcal{C}$ and $\mathcal{D}$ be abelian A-categories, and let $T = (T^i)_{i \in \mathbb{N}}$ be an exact A-linear ∂-functor from $\mathcal{C}$ to $\mathcal{D}$. Assume:

(i) for every noetherian object M of $\mathcal{C}$ killed by a power of J, and every n, the object $T^n(M)$ is noetherian;

(ii) for every finitely generated graded A-algebra B killed by J, and every noetherian graded $(B, \mathcal{C})$-module N, the graded $(B, \mathcal{D})$-modules $T^n(N)$ are noetherian.

Then, for every AR-J-adic noetherian object X of $\underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})$, the objects $T^n(X)$ are AR-J-adic and noetherian.

Since the T^n commute with translations, it suffices to treat the case where X is J-adic noetherian. Then $\mathrm{gr}_J(X)$ is a noetherian graded $\mathrm{gr}_J(A)$-module. By hypothesis (ii) and a variant of a corollary of Shih's theorem (Appendix A.3), for each n one has: $T^n(X)$ satisfies (MLAR), and $\mathrm{gr}_J T^n(X)$ is a noetherian graded $(\mathrm{gr}_J(A), \mathcal{D})$-module. The Artin–Rees lemma 4.2.6 completes the proof.

It follows that the ∂-functor

$$T = (T^i)_{i \in \mathbb{N}} : \underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C}) o \underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{D})$$

induces an exact A-linear ∂-functor

$$\mathcal{E}_{\mathcal{C}} \longrightarrow \mathcal{E}_{\mathcal{D}},$$

commuting with translations, and hence, after passing to the quotient, an exact A-linear ∂-functor, again denoted (T^i),

$$\text{AR-}J\text{-adn}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adn}(\mathcal{D}).$$

Appendix. Shih's theorem

The purpose of this appendix is to prove Shih's theorem in the form used in the proof of 5.3.1. Since this form is more general than the one stated in EGA $\mathcal{O}_{\mathrm{III}}$, a direct proof, adapted from that of loc. cit., is required.

Fix an integer $r_0 \geq 1$. The data, for every $r \geq r_0$, of a bigraded object $E_r = (E_r^{pq})_{p,q \in \mathbb{Z}}$, endowed with a differential d_r of bidegree $(r, 1-r)$, together with isomorphisms

$$H(E_r) \xrightarrow{\sim} E_{r+1},$$

will be called a *spectral filter*. The notion of morphism of spectral filters is evident.

A.1. Spectral sequences attached to ∂-functors

Let $\mathcal{C}$ be an abelian category and let X be an object of $\mathcal{C}$ with a decreasing filtration $(F^pX)_{p\in\mathbb{Z}}$. For every exact ∂-functor from $\mathcal{C}$ to an abelian category $\mathcal{D}$, the method of exact couples gives a spectral sequence

$$\mathbb{E}(X) : E_1^{pq} = T^{p+q}(F^pX/F^{p+1}X) \Rightarrow E^n = T^n(X),$$

whose filtration on E^n is

$$F^p(E^n) = \mathrm{Im}(T^n(F^pX) \to T^n(X)) = \mathrm{Ker}(T^n(X) \to T^n(X/F^pX)).$$

If the filtration on X is finite, this spectral sequence is weakly convergent (EGA $\mathcal{O}_{\mathrm{III}}$, 11.1.3).

The construction is functorial. If $f : X \to Y$ is a morphism of filtered objects, there is a morphism of spectral sequences

$$\mathbb{E}(f) : \mathbb{E}(X) \to \mathbb{E}(Y)$$

which is $T^{p+q}(\mathrm{gr}^p f)$ on E_1^{pq} and $T^n(f)$ on the abutment. If $u : X \to X$ is an endomorphism with $u(F^pX) \subset F^{p+h}X$ for an integer h independent of p, then $\mathbb{E}(u)$ is an endomorphism of bidegree $(h, -h)$ of the spectral filter, commuting with the differentials.

A.2. The case of projective systems

Let $X = (X_n)_{n\in\mathbb{Z}}$ be a projective system of objects of $\mathcal{C}$, with $X_i = 0$ for $i < 0$. If T is an exact ∂-functor from $\mathcal{C}$ to $\mathcal{D}$, it extends naturally to a ∂-functor on $\underline{\mathrm{Hom}}(\mathbb{Z}^\circ, \mathcal{C})$. Giving X its natural filtration as a projective system yields a spectral sequence

$$\mathbb{E}(X) = \mathbb{E}(X_n)_{n\in\mathbb{Z}},$$

where $\mathbb{E}(X_n)$ is the spectral sequence associated with T and the induced filtration on X_n.

If X is strict, the filtration of the abutment $E^n(X)$ in this spectral sequence is the natural filtration of the projective system. Indeed,

$$F^p(E^n(X)) = \mathrm{Im}(T^n(F^pX) \to T^n(X)),$$

and $(F^pX)_k = X_k$ for $k \geq p$ and 0 for $k < p$.

Suppose a ring S acts on X compatibly with its natural filtration. Then, for each $r \geq 1$, the bigraded object $E_r(X)$ is canonically a graded $(\mathrm{gr}^\bullet S, \mathcal{D})$-module; the same holds for

$$\mathcal{E}_r^{(n)}(X) = (\mathcal{E}_r^{p,q}(X))_{p+q=n}.$$

Moreover S acts on $T^n(X)$, and whenever $T^n(X)$ satisfies the Mittag-Leffler condition, the strict graded object $\mathrm{gr}_S^\bullet T^n(X)$ is a graded $(\mathrm{gr}^\bullet S, \mathcal{D})$-module.

Theorem A.3. Suppose that, for an integer n, the graded $(\mathrm{gr}^\bullet S, \mathcal{D})$-modules

$$\mathcal{E}_1^{(n)}(X) = T^n(\mathrm{gr}_S^\bullet X), \qquad \mathcal{E}_1^{(n-1)}(X) = T^{n-1}(\mathrm{gr}_S^\bullet X)$$

are noetherian. Then:

(i) $T^n(X)$ satisfies (MLAR);

(ii) $\mathrm{gr}_S^\bullet T^n(X)$ is a noetherian graded $(\mathrm{gr}^\bullet S, \mathcal{D})$-module.

For every pair (m, r) one has

$$\mathcal{E}_r^{(m)}(X) = (\mathcal{E}_r^{p,m-p}(X))_{p\in\mathbb{N}},$$

a graded submodule of $\mathcal{E}_1^{(m)}(X)$. Thus, for fixed m, the $\mathcal{E}_r^{(m)}(X)$ form an increasing sequence indexed by r. Since $\mathcal{E}_1^{(n)}(X)$ and $\mathcal{E}_1^{(n-1)}(X)$ are noetherian, the corresponding sequences are stationary. Hence for some r_0 and all $r \geq r_0$,

$$\mathcal{E}_r^{(n)}(X) = \mathcal{E}_{r_0}^{(n)}(X), \qquad \mathcal{E}_r^{(n-1)}(X) = \mathcal{E}_{r_0}^{(n-1)}(X).$$

An equivalent characterization of the Mittag-Leffler condition then shows that the spectral sequence $\mathbb{E}(X)$ is weakly convergent in degrees n and $n-1$. It follows that the filtration of $E^n(X) = T^n(X)$ is separated and that $\mathcal{E}_\infty^{(n)}(X)$ is a subobject of the noetherian module $\mathcal{E}_{r_0}^{(n)}(X)$.

To prove that $T^n(X)$ satisfies (ML), it is enough to prove that for each p the sequence

$$\mathrm{Im}(T^n(F^pX) \to T^n(X/F^{p+r}X))_{r\geq 1}$$

is stationary. Weak convergence identifies these images, for r large, with the kernels of maps to $E_\infty^{(n)}(X)$; since these kernels lie in a noetherian object, they are stationary. Assertion (ii) follows because $\mathrm{gr}_S^\bullet T^n(X)$ is a subquotient of $\mathcal{E}_\infty^{(n)}(X)$.

A.4. Application to ∂-functors

Let $\mathcal{C}$ and $\mathcal{D}$ be abelian A-categories, and let $T = (T^i)_{i \in \mathbb{N}}$ be an exact A-linear ∂-functor from $\mathcal{C}$ to $\mathcal{D}$. Let X be an object of $\underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathcal{C})$, and let S be a graded A-algebra acting on X compatibly with the transition maps. Assume that X is AR-J-adic noetherian; that $T^n(M)$ is noetherian for each noetherian M killed by a power of J; and that, for every finitely generated graded A-algebra B killed by J and every noetherian graded $(B, \mathcal{C})$-module N, the graded $(B, \mathcal{D})$-modules $T^n(N)$ are noetherian.

Theorem A.4.1. Under these hypotheses, the objects $T^n(X)$ are AR-J-adic and noetherian.

By translating one is reduced to the case where X is J-adic noetherian. Then $\mathrm{gr}^\bullet_J(X)$ is a noetherian graded $(\mathrm{gr}^\bullet_J(A), \mathcal{C})$-module. Hypothesis (iii) makes $\mathcal{E}_1^{(n)}(X)$ and $\mathcal{E}_1^{(n-1)}(X)$ noetherian, and Theorem A.3 gives the conclusion. This completes Exposé V.

Exposé VI. ℓ-adic cohomology

1. Categories of constructible ℓ-adic sheaves

1.1. Generalities

Let X be a scheme and let ℓ be a prime number. We write

$$\mathbb{Z}_\ell\text{-fc}(X)$$

for the category of constructible $\mathbb{Z}_\ell$-sheaves on X, defined in SGA 5, V. It is an abelian category. We write

$$\mathbb{Q}_\ell\text{-fc}(X)$$

for the category of constructible $\mathbb{Q}_\ell$-sheaves, i.e. the quotient of the preceding category by the thick subcategory of torsion sheaves. We have also defined the categories

$$\text{AR-}\mathbb{Z}_\ell\text{-fc}(X), \qquad \text{AR-}\mathbb{Q}_\ell\text{-fc}(X).$$

Recall that giving a constructible $\mathbb{Z}_\ell$-sheaf $\mathcal{F}$ is equivalent to giving a projective system $(\mathcal{F}_n)_{n\in\mathbb{N}}$ of constructible $\mathbb{Z}/\ell^{n+1}\mathbb{Z}$-modules on X such that, for every n, the transition morphism $\mathcal{F}_{n+1} \to \mathcal{F}_n$ factors through an isomorphism

$$\mathcal{F}_{n+1}/\ell^{n+1}\mathcal{F}_{n+1} \xrightarrow{\sim} \mathcal{F}_n.$$

For two such sheaves $\mathcal{F} = (\mathcal{F}_n)$ and $\mathcal{G} = (\mathcal{G}_n)$,

$$\mathrm{Hom}(\mathcal{F}, \mathcal{G}) = \varprojlim_n \mathrm{Hom}(\mathcal{F}_n, \mathcal{G}_n).$$

For each n, the canonical map

$$\mathrm{Hom}(\mathcal{F}, \mathcal{G}) \longrightarrow \mathrm{Hom}(\mathcal{F}_n, \mathcal{G}_n)$$

is surjective, and its kernel consists of morphisms which factor through a torsion object.

The ring $\mathbb{Z}_\ell$ acts in the evident way on every component of a constructible ℓ-adic sheaf, and hence on the sheaf itself. Every morphism is compatible with these actions. Thus the category of constructible ℓ-adic sheaves is canonically an abelian $\mathbb{Z}_\ell$-category. We shall call these objects constructible $\mathbb{Z}_\ell$-sheaves. This should not be confused with a module over the sheaf of rings $(\mathbb{Z}_\ell)_X$, a notion not used here. The category will also be denoted $\mathbb{Z}_\ell$-fc(X).

1.2. Twisted constant constructible $\mathbb{Z}_\ell$-sheaves

A constructible $\mathbb{Z}_\ell$-sheaf

$$\mathcal{F} = (\mathcal{F}_n)_{n\in\mathbb{N}}$$

is called *twisted constant* if and only if all the components $\mathcal{F}_n$ are locally constant on X_{et}. The category of twisted constant constructible $\mathbb{Z}_\ell$-sheaves is the full subcategory of $\mathbb{Z}_\ell$-fc(X) generated by these objects.

Assume that ℓ is invertible on X. For each n, raising to the ℓth power gives an endomorphism of $\mathbb{G}_m$, and the sheaf

$$\mathbb{Z}/\ell^{n+1}\mathbb{Z}(1)$$

is the sheaf $\mu_{\ell^{n+1}}$ of ℓ^{n+1}st roots of unity. The projective system of these sheaves defines a twisted constant $\mathbb{Z}_\ell$-sheaf denoted $\mathbb{Z}_\ell(1)$.

Let $\mathcal{F} = (\mathcal{F}_n)$ be a twisted constant constructible $\mathbb{Z}_\ell$-sheaf. For every geometric point $\bar{x}$ of X, the geometric fiber $\mathcal{F}_{\bar{x}}$ is a finite type $\mathbb{Z}_\ell$-module endowed with a continuous action of the fundamental group $\pi_1(X, \bar{x})$. The fiber functor

$$\mathcal{F} \longmapsto \mathcal{F}_{\bar{x}}$$

is an equivalence between twisted constant constructible $\mathbb{Z}_\ell$-sheaves on X and finite type $\mathbb{Z}_\ell$-modules with continuous action of $\pi_1(X, \bar{x})$.

For a finite type $\mathbb{Z}_\ell$-module M we write M_X for the corresponding twisted constant sheaf. In particular,

$$\mathbb{Z}_\ell(r) = \mathbb{Z}_\ell(1)^{\otimes r} \qquad (r \in \mathbb{Z}),$$

where for $r < 0$ one sets $\mathbb{Z}_\ell(r) = \mathbb{Z}_\ell(-r)^\vee$. If $\mathcal{F}$ is a twisted constant $\mathbb{Z}_\ell$-sheaf, define

$$H^i(X, \mathcal{F}) = \varprojlim_n H^i(X, \mathcal{F}_n).$$

These groups are finite type $\mathbb{Z}_\ell$-modules when X is proper over a separably closed field.

1.3. Invertible ℓ-adic sheaves

Let $\mathcal{L}$ be a constructible $\mathbb{Z}_\ell$-sheaf. We say that $\mathcal{L}$ is *invertible* if, for every n, the component $\mathcal{L}_n$ is an invertible $(\mathbb{Z}/\ell^{n+1}\mathbb{Z})_X$-module, i.e. locally free of rank one. Equivalently, $\mathcal{L}$ is locally free in the sense of the preceding paragraph and the geometric fibers of the $\mathcal{L}_n$ are non-zero cyclic abelian groups.

For an invertible $\mathcal{L}$ put
$$\mathcal{L}^{-1} = \mathcal{H}om(\mathcal{L}, \mathbb{Z}_\ell).$$
There is then an isomorphism
$$\mathcal{L} \otimes \mathcal{L}^{-1} \xrightarrow{\sim} \mathbb{Z}_\ell,$$
obtained componentwise. Following EGA $\mathcal{O}_I$, 5.4.4, set $\mathcal{L}^{\otimes 0} = \mathbb{Z}_\ell$ and, for $n \geq 1$, $\mathcal{L}^{\otimes(-n)} = (\mathcal{L}^{-1})^{\otimes n}$. With this notation there are canonical functorial isomorphisms
$$\mathcal{L}^{\otimes m} \otimes \mathcal{L}^{\otimes n} \xrightarrow{\sim} \mathcal{L}^{\otimes(m+n)}$$
for all integers m, n, satisfying the usual associativity rules. Thus $(\mathcal{L}^{\otimes i})_{i\in\mathbb{N}}$ and $(\mathcal{L}^{\otimes i})_{i\in\mathbb{Z}}$ are graded $\mathbb{Z}_\ell$-algebras. The sheaf $\mathbb{Z}_\ell(1)$ is invertible, and the convention above gives $\mathbb{Z}_\ell(r) = \mathbb{Z}_\ell(1)^{\otimes r}$ for all $r \in \mathbb{Z}$.

1.4. Categories built from constructible $\mathbb{Z}_\ell$-sheaves

Let A be a finite $\mathbb{Z}_\ell$-algebra. Define the category
$$A\text{-fc}(X)$$
of constructible A-sheaves as follows. An object is a pair $(\mathcal{F}, u)$ consisting of a constructible $\mathbb{Z}_\ell$-sheaf $\mathcal{F}$ and a homomorphism of $\mathbb{Z}_\ell$-algebras
$$u : A \longrightarrow \mathrm{End}(\mathcal{F}).$$
A morphism $(\mathcal{F}, u) \to (\mathcal{F}', u')$ is a morphism $f : \mathcal{F} \to \mathcal{F}'$ of constructible $\mathbb{Z}_\ell$-sheaves such that, for every $a \in A$, the square
$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{f} & \mathcal{F}' \\ \downarrow{\scriptstyle u(a)} & & \downarrow{\scriptstyle u'(a)} \\ \mathcal{F} & \xrightarrow{f} & \mathcal{F}' \end{array}$$
commutes. The category A-fc(X) is abelian, and the forgetful functor to $\mathbb{Z}_\ell$-fc(X) is exact and faithful. The dual of a constructible A-sheaf $\mathcal{F}$ is
$$\mathcal{F}^\vee = \mathcal{H}om(\mathcal{F}, A_X).$$
If $f : X \to Y$ is a morphism of schemes, inverse image and direct image extend in the evident way to the corresponding categories of constructible A-sheaves.

Let $p : \mathbb{Z}_\ell\text{-fc}(X) \to \mathbb{Q}_\ell\text{-fc}(X)$ be the quotient functor. For a constructible $\mathbb{Z}_\ell$-sheaf $\mathcal{F}$, the functor obtained by composing p with $\mathcal{G} \mapsto \mathcal{F} \otimes \mathcal{G}$ makes multiplication by non-zero elements of $\mathbb{Z}_\ell$ invertible, and hence defines a right-exact functor

$$\mathbb{Q}_\ell\text{-fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-fc}(X).$$

Varying the first argument as well gives a right-exact bifunctor, called tensor product,

$$\mathbb{Q}_\ell\text{-fc}(X) \times \mathbb{Q}_\ell\text{-fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-fc}(X).$$

If X is connected and $\mathcal{F}, \mathcal{G}$ are twisted constant constructible $\mathbb{Q}_\ell$-sheaves corresponding to representations V, W of $\pi_1(X)$, then $\mathcal{F} \otimes \mathcal{G}$ is again locally constant and corresponds to the diagonal representation on $V \otimes W$.

1.5. Constructible ℓ-adic sheaves over the spectrum of a field

If k is separably closed, then $\mathbb{Z}_\ell\text{-fc}(\operatorname{Spec} k)$ is identified with the category of finite type $\mathbb{Z}_\ell$-modules, and $\mathbb{Q}_\ell\text{-fc}(\operatorname{Spec} k)$ with finite-dimensional $\mathbb{Q}_\ell$-vector spaces.

If k is a field and $\bar{k}$ is a separable closure, the geometric-fiber functor at $\operatorname{Spec}(\bar{k}) \to \operatorname{Spec}(k)$ gives an equivalence

$$\mathbb{Z}_\ell\text{-fc}(\operatorname{Spec} k) \xrightarrow{\sim} \{\text{finite type } \mathbb{Z}_\ell\text{-modules with continuous action of } \operatorname{Gal}(\bar{k}/k)\}.$$

After inverting torsion, this identifies $\mathbb{Q}_\ell\text{-fc}(\operatorname{Spec} k)$ with the category of finite-dimensional continuous ℓ-adic representations of $\operatorname{Gal}(\bar{k}/k)$.

If k is finite with q elements, let

$$F_k \in \operatorname{Gal}(\bar{k}/k)$$

denote the geometric Frobenius. A continuous ℓ-adic representation of $\operatorname{Gal}(\bar{k}/k)$ is determined by the action of F_k. If V is a finite-dimensional $\mathbb{Q}_\ell$-vector space with continuous action of $\operatorname{Gal}(\bar{k}/k)$, the eigenvalues of F_k on V are algebraic numbers whose inverses are integral over $\mathbb{Z}$.

Let

$$\mathbb{Z}_\ell\text{-modn}(\pi_1)$$

be the category of finite type $\mathbb{Z}_\ell$-modules endowed with a continuous action of π_1.

Lemma 1.2.4.2. The preceding projective-limit functor

$$\varprojlim_{\mathbb{N}} : (\ell\text{-}\mathbb{Z})\text{-ad}(\mathrm{Abf}(\pi_1)) \longrightarrow \mathbb{Z}_\ell\text{-modn}(\pi_1)$$

is an equivalence of categories.

It is clearly essentially surjective: a finite type $\mathbb{Z}_\ell$-module E with a continuous action of π_1 is obtained from the projective system

$$(E/\ell^{n+1}E)_{n\in\mathbb{N}}$$

with the evident actions of π_1.

2. Formalism of ℓ-adic cohomology

2.1. Inverse image

Let $f : X \to Y$ be a morphism of locally noetherian preschemes. The inverse-image functor

$$f^* : \mathrm{Ab}(Y) \longrightarrow \mathrm{Ab}(X)$$

is exact and therefore extends in the evident way to an exact functor, denoted by the same symbol,

$$f^* : \underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathrm{Ab}(Y)) \longrightarrow \underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathrm{Ab}(X)).$$

It plainly carries constructible $\mathbb{Z}_\ell$-sheaves to constructible $\mathbb{Z}_\ell$-sheaves and locally AR-zero systems to locally AR-zero systems. With the notation of 1.5.4 it therefore induces inverse-image functors

$$\begin{aligned} f^* : \mathbb{Z}_\ell\text{-fc}(Y) &\longrightarrow \mathbb{Z}_\ell\text{-fc}(X), \\ f^* : \mathbb{Q}_\ell\text{-fc}(Y) &\longrightarrow \mathbb{Q}_\ell\text{-fc}(X). \end{aligned}$$

These functors are exact, commute with tensor products, and preserve twisted constant sheaves. If A is a finite $\mathbb{Z}_\ell$-algebra, the same construction gives inverse-image functors for constructible A-sheaves.

2.2. Direct image

Let again $f : X \to Y$ be a morphism of locally noetherian preschemes. The direct-image functor on étale sheaves is left exact and so has right derived functors. Applying it term by term to a projective system gives a left exact functor

$$f_* : \underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathrm{Ab}(X)) \longrightarrow \underline{\mathrm{Hom}}(\mathbb{N}^\circ, \mathrm{Ab}(Y)).$$

The corresponding derived functors are denoted $R^i f_*$. When f is proper and of finite type, the finiteness theorem for étale cohomology shows that $R^i f_*$ carries constructible $\mathbb{Z}_\ell$-sheaves to constructible $\mathbb{Z}_\ell$-sheaves. Thus one obtains functors

$$R^i f_* : \mathbb{Z}_\ell\text{-fc}(X) \longrightarrow \mathbb{Z}_\ell\text{-fc}(Y)$$

and, after passage to the quotient by torsion,

$$R^i f_* : \mathbb{Q}_\ell\text{-fc}(X) \longrightarrow \mathbb{Q}_\ell\text{-fc}(Y).$$

They are compatible with formation of fibers in the usual sense: at a geometric point $\bar{y}$ of Y, the fiber of $R^i f_* \mathcal{F}$ is the ith cohomology of the geometric fiber $X_{\bar{y}}$ with coefficients in the pullback of $\mathcal{F}$.

2.3. Tensor products and internal Hom

The tensor product of projective systems is defined level by level. If $\mathcal{F} = (\mathcal{F}_n)$ and $\mathcal{G} = (\mathcal{G}_n)$ are constructible $\mathbb{Z}_\ell$-sheaves, set

$$(\mathcal{F} \otimes \mathcal{G})_n = \mathcal{F}_n \otimes_{\mathbb{Z}/\ell^{n+1}\mathbb{Z}} \mathcal{G}_n.$$

This again defines a constructible $\mathbb{Z}_\ell$-sheaf. The functor is right exact in each variable; its left derived functors are denoted $\mathrm{Tor}_i^{\mathbb{Z}_\ell}(\mathcal{F}, \mathcal{G})$. For constructible $\mathbb{Q}_\ell$-sheaves the tensor product is exact.

The internal Hom is defined similarly by

$$\mathcal{H}om(\mathcal{F}, \mathcal{G})_n = \mathcal{H}om_{\mathbb{Z}/\ell^{n+1}\mathbb{Z}}(\mathcal{F}_n, \mathcal{G}_n),$$

with the transition morphisms induced in the evident way. Its right derived functors are written $\mathcal{E}xt^i(\mathcal{F}, \mathcal{G})$. The familiar adjunctions between tensor product and internal Hom remain valid in the ℓ-adic setting.

2.4. Cohomology

For a locally noetherian scheme X and a constructible $\mathbb{Z}_\ell$-sheaf $\mathcal{F} = (\mathcal{F}_n)$, define

$$H^i(X, \mathcal{F}) = \varprojlim_n H^i(X, \mathcal{F}_n).$$

When X is proper over a separably closed field, these groups are finite $\mathbb{Z}_\ell$-modules. Passing to $\mathbb{Q}_\ell$ gives

$$H^i(X, \mathcal{F} \otimes \mathbb{Q}_\ell) = H^i(X, \mathcal{F}) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

If X is not proper, one uses cohomology with proper supports, defined by compactification. For separated morphisms of finite type, the construction is independent of the chosen compactification.

2.5. Finite cohomological dimension

Let X be of finite type over a separably closed field and let ℓ be invertible on X. Then X has finite ℓ-cohomological dimension. More precisely, if $\dim X = d$, then for every torsion constructible sheaf F killed by a power of ℓ, one has

$$H^i(X, F) = 0 \qquad (i > 2d),$$

and similarly for cohomology with proper supports. This is the usual cohomological dimension theorem of SGA 4, XIV.

2.6. Cohomology with proper supports

Let $f : X \to S$ be a separated morphism of finite type between locally noetherian schemes, and let $\mathcal{F}$ be a constructible $\mathbb{Z}_\ell$-sheaf on X. The cohomology with proper supports

$$H^i_c(X/S, \mathcal{F})$$

is the constructible $\mathbb{Z}_\ell$-sheaf on S obtained by compactifying f and applying the corresponding direct image with extension by zero. If $j : U \to X$ is an open immersion and $i : Z \to X$ is the complementary closed immersion, then for every constructible $\mathbb{Z}_\ell$-sheaf $\mathcal{F}$ on U there is the usual long exact localization sequence

$$\cdots \to R^i f_!(j_!\mathcal{F}) \to R^i f_!(Rj_*\mathcal{F}) \to R^i f_!(i_* i^* Rj_*\mathcal{F}) \to \cdots .$$

3. ℓ-adic L- and Z-functions

3.1. Definitions

Let $\mathbb{F}_q$ be a finite field with q elements and let X be a scheme of finite type over $\mathbb{F}_q$. Let $|X|$ be the set of closed points of X. For $x \in |X|$, write $d(x)$ for the degree of the residue field $k(x)$ over $\mathbb{F}_q$, and put $N(x) = q^{d(x)}$.

Let $\mathcal{F}$ be a constructible $\mathbb{Q}_\ell$-sheaf on X. For each closed point x, the geometric fiber $\mathcal{F}_{\bar{x}}$ is a finite-dimensional $\mathbb{Q}_\ell$-vector space with a continuous action of $\mathrm{Gal}(\overline{k(x)}/k(x))$. Let F_x be geometric Frobenius. If the eigenvalues of F_x on $\mathcal{F}_{\bar{x}}$ are $\alpha_1, \ldots, \alpha_n$, then

$$\det(1 - F_x t^{d(x)} \mid \mathcal{F}_{\bar{x}}) = \prod_{i=1}^{n} (1 - \alpha_i t^{d(x)}).$$

The associated Z-function is

$$Z(X, \mathcal{F}; t) = \prod_{x \in |X|} \frac{1}{\det(1 - F_x t^{d(x)} \mid \mathcal{F}_{\bar{x}})} \in \mathbb{Q}_\ell[[t]].$$

For $\mathcal{F} = \mathbb{Q}_\ell$ one writes simply $Z(X; t)$.

3.2. Lefschetz trace formula

Theorem 3.2.1. Lefschetz trace formula. Let X be proper and smooth over $\mathbb{F}_q$ and let $\mathcal{F}$ be a constructible $\mathbb{Q}_\ell$-sheaf. Then

$$Z(X, \mathcal{F}; t) = \prod_{i=0}^{2 \dim X} \det(1 - F^* t \mid H^i(X \otimes \overline{\mathbb{F}}_q, \mathcal{F}))^{(-1)^{i+1}}.$$

The proof uses the Lefschetz formula for correspondences on proper smooth varieties, as developed in SGA 5, III, together with the Artin–Verdier theory of the L-functor. The essential point is that the global trace of Frobenius on cohomology is equal to the sum of its local terms at the fixed points, which are the rational points over finite extensions of $\mathbb{F}_q$.

3.3. The Weil conjectures

Let X be proper and smooth over a finite field $\mathbb{F}_q$, and let ℓ be different from the characteristic of $\mathbb{F}_q$. The Weil conjectures assert:

(i) the characteristic polynomials

$$\det(1 - F^* t \mid H^i(X \otimes \overline{\mathbb{F}}_q, \mathbb{Q}_\ell))$$

have integral coefficients and are independent of ℓ;

(ii) the eigenvalues of F^* on $H^i(X \otimes \overline{\mathbb{F}}_q, \mathbb{Q}_\ell)$ are algebraic integers all of whose complex absolute values are $q^{i/2}$;

(iii) Poincaré duality gives perfect pairings

$$H^i(X \otimes \overline{\mathbb{F}}_q, \mathbb{Q}_\ell) \times H^{2d-i}(X \otimes \overline{\mathbb{F}}_q, \mathbb{Q}_\ell(d)) \longrightarrow \mathbb{Q}_\ell.$$

These conjectures are proved by Deligne. The proof uses the trace formula and the theory of weights.

4. Cohomology of ℓ-adic completions

Let X be proper over $\mathrm{Spec}(\mathbb{Z}[1/\ell])$. For each $n \geq 1$, put

$$X_n = X \otimes_{\mathbb{Z}[1/\ell]} \mathbb{Z}[1/\ell, \zeta_{\ell^n}],$$

where ζ_{ℓ^n} is a primitive ℓ^nth root of unity. The cohomology groups

$$H^i(X_n, \mathbb{Z}/\ell^n\mathbb{Z})$$

form a projective system, and its limit

$$H^i_{\ell\text{-comp}}(X) = \varprojlim_n H^i(X_n, \mathbb{Z}/\ell^n\mathbb{Z})$$

is called the ith completed ℓ-adic cohomology group of X. These are finite type $\mathbb{Z}_\ell$-modules and coincide, in the proper smooth case, with the usual ℓ-adic cohomology groups. With coefficients in a constructible $\mathbb{Z}_\ell$-sheaf $\mathcal{F}$, define similarly

$$H^i_{\ell\text{-comp}}(X, \mathcal{F}) = \varprojlim_n H^i(X_n, \mathcal{F}/\ell^n\mathcal{F}).$$

There is also an Iwasawa spectral sequence. If $G = \mathrm{Gal}(\mathbb{Q}(\zeta_{\ell^\infty})/\mathbb{Q}) \cong \mathbb{Z}_\ell^\times$, then G acts on $H^i_{\ell\text{-comp}}(X)$ by transport of structure. These groups may therefore be regarded as modules over the Iwasawa algebra

$$\Lambda = \mathbb{Z}_\ell[[G]] = \varprojlim_n \mathbb{Z}_\ell[G/\ell^n\mathbb{Z}_\ell^\times].$$

The resulting Λ-modules $H^i_{\ell\text{-comp}}(X)$ are of finite type. The proof uses the Hochschild–Serre spectral sequence for the covers $X_n \to X$ and the finiteness theorem for cohomology of proper schemes.

5. Applications

Theorem 5.1.1. Rationality of Z-functions. Let X be of finite type over $\mathbb{F}_q$ and let $\mathcal{F}$ be a constructible $\mathbb{Q}_\ell$-sheaf on X. Then $Z(X, \mathcal{F}; t)$ is a rational function of t.

When X is proper and smooth this follows from the Lefschetz trace formula 3.2.1. The general case is deduced by dévissage, stratification, and the long exact cohomology sequence.

Theorem 5.2.1. Unit theorem. Let K be a number field, let $\mathcal{O}_K$ be its ring of integers, and let ℓ be a prime. Then $\mathcal{O}_K^\times$ is finitely generated, and its rank is

$$\mathrm{rank}(\mathcal{O}_K^\times) = r_1 + r_2 - 1,$$

where r_1 is the number of real places and r_2 the number of complex places of K.

This classical theorem can be interpreted in terms of ℓ-adic cohomology. The Iwasawa spectral sequence for $\mathrm{Spec}(\mathcal{O}_K[1/\ell])$ gives a description of the unit group in terms of Iwasawa modules.

Theorem 5.3.1. Galois structure of cohomology. Let k be a number field and let X be a smooth projective variety over k. Let $\bar{k}$ be an algebraic closure of k. Then the representation of $\mathrm{Gal}(\bar{k}/k)$ on

$$H^i_{\mathrm{et}}(X \otimes_k \bar{k}, \mathbb{Q}_\ell)$$

is semisimple, and its composition factors are independent of ℓ for ℓ different from the residual characteristics.

The semisimplicity follows from Deligne's theory of weights. The independence of ℓ is a consequence of the Weil conjectures and the theory of algebraic cycles. This completes Exposé VI.

Continuation anchor: SGA 5, pages 301–350, continuation around source line 5467.

SGA 5 – Batch 028, continuation reader

Exposé VII continuation through Exposé XV

Working English LaTeX translation

This reader continues after the bridge file in this package. It translates the source range from the Exposé VII Chern-class continuation through the end of the current SGA 5 working source, including Exposés VIII, XI, XII, and XV.

Exposé VII – Chern Classes, Self-Intersection, Flag Schemes, and Blow-Ups

The source block available here resumes in the middle of the proof of the projective bundle formula, at the continuation of 2.2.2. The translation follows that source block and then continues through the end of the exposé.

2.2. Projective bundles: continuation

Let $p : P \to S$ be the projective bundle associated with a locally free $\mathcal{O}_S$-module E of rank $r+1$, and let

$$\xi = c_1(\mathcal{O}_P(1)) \in H^2(P, \mu_n).$$

The remaining point in the proof of the projective bundle theorem is to show that, for $0 \leq i \leq r$, the class ξ^i is a free element over the cohomology ring of S. It is enough to verify this for ξ^r, since the lower powers are then handled by the usual triangular argument. This is exactly the compatibility of the trace morphism with the fundamental class in the diagram

$$\begin{array}{ccc} H^{2r}(P, \mu_n^{\otimes r}) & \xrightarrow{\mathrm{Tr}} & A \\ \cup p^*(x) \uparrow & & \uparrow x \\ H^0(P, \mathbb{Z}/n\mathbb{Z}) & \xrightarrow{\sim} & A, \end{array}$$

where p^* is induced by a closed point of the fiber.

Corollary 2.2.4. Let L be a complex of A-modules. The morphism

$$R\Gamma_*(S, L) \longrightarrow R\Gamma_*(P, p^*L)$$

induced by the canonical splitting of the projective bundle formula is an isomorphism after applying the corresponding projector.

Indeed, it suffices to apply $R\Gamma_*$ to the isomorphism furnished by the projective bundle formula.

For later use set

$$A^*(X) = \bigoplus_i H^{2i}(X, \mu_n^{\otimes i}).$$

This notation will be used systematically.

Corollary 2.2.6. The pullback homomorphism

$$p^* : A^*(S) \longrightarrow A^*(P)$$

is an injective homomorphism of unital rings, and $1, \xi, \ldots, \xi^r$ form a basis of the $A^*(S)$-module $A^*(P)$. In particular $A^*(P)$ is free over $A^*(S)$ of rank equal to the rank of E.

This is the form of the projective bundle theorem used in the construction of Chern classes and in the splitting principle below.

3. Chern classes

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module of constant rank r. Put $P = \mathbb{P}(E)$ and let $\xi = c_1(\mathcal{O}_P(1))$. The projective bundle formula shows that there is a unique relation

$$\xi^r - c_1\xi^{r-1} + c_2\xi^{r-2} - \cdots + (-1)^r c_r = 0,$$

with

$$c_i \in H^{2i}(S, \mu_n^{\otimes i}).$$

If E has non-constant rank, the scheme S is partitioned into open-and-closed subsets S_r on which E has rank r. Since cohomology decomposes over this disjoint union, the preceding construction defines the classes on each component and hence on S.

Definition 3.2. For a locally free $\mathcal{O}_S$-module E, the element $c_i(E) \in H^{2i}(S, \mu_n^{\otimes i})$ obtained by the preceding construction is called the ith Chern class of E. If the rank of E is bounded, the total Chern class is

$$c(E) = \sum_i c_i(E) \in A^*(S).$$

It is immediate from the definition that $c_0(E) = 1$, and if E has rank at most d, then $c_i(E) = 0$ for $i > d$.

Proposition 3.4. The Chern classes satisfy the following properties.

(i) Normalization: if L is invertible and $[L]$ is the image of the class of L under the boundary map

$$H^1(S, \mathcal{O}_S^*) \to H^2(S, \mu_n),$$

then $c(L) = 1 + [L]$.

(ii) Base change: for $f : S' \to S$, one has $f^*c_i(E) = c_i(f^*E)$.

(iii) Additivity: for an exact sequence of locally free modules

$$0 \to E' \to E \to E'' \to 0,$$

one has

$$c_n(E) = \sum_{i+j=n} c_i(E')c_j(E'').$$

Proof. For line bundles the projective bundle is S itself and $\mathcal{O}_P(1)$ is identified with the dual of the given line bundle, giving the normalization. Base change follows immediately from the functoriality of $\mathbb{P}(E)$, of $\mathcal{O}(1)$, and of the defining relation.

For additivity one first reduces, by partitioning S into open-and-closed pieces, to the case where all ranks are constant. Then one applies the splitting principle. Let $h : Z \to S$ be the product of the flag schemes of E' and E''. The pullbacks h^*E' and h^*E'' have finite filtrations with invertible quotients, say L_i and M_j. The pullback h^*E has a filtration with consecutive quotients the L_i and M_j. The lemma below gives

$$c(h^*E) = \prod_i(1 + [L_i])\prod_j(1 + [M_j]) = c(h^*E')c(h^*E'').$$

The morphism h is a composite of projective-bundle morphisms, so $h^* : A^*(S) \to A^*(Z)$ is injective by Corollary 2.2.6. The desired identity follows.

Lemma 3.6. Let E be locally free and endowed with a finite filtration

$$0 = E_0 \subset E_1 \subset \cdots \subset E_r = E$$

whose successive quotients $L_i = E_i/E_{i-1}$ are invertible. Then

$$c(E) = \prod_{i=1}^{r}(1 + [L_i]).$$

The proof is the standard flag argument. On $P = \mathbb{P}(E)$ the filtration defines closed subschemes

$$S \simeq Y_1 \subset Y_2 \subset \cdots \subset Y_r = P.$$

The immersion $Y_i \hookrightarrow Y_{i+1}$ is defined by an invertible ideal identified with

$$p^*(L_i^\vee) \otimes \mathcal{O}_{Y_{i+1}}(-1).$$

Using the projection formula for the Gysin morphism and the identity $u_*(1_Z) = -[J]$ for an immersion defined by an invertible ideal J, one proves by induction the vanishing relation

$$(a_i)_* a_i^* iggl(\prod_{j \leq i} p^*(L_j^\vee) \otimes \mathcal{O}_P(-1)\biggr) = 0.$$

For $i = r$ this is precisely

$$\prod_{j=1}^{r} (\xi + c_1(p^* L_j)) = 0,$$

which is the desired expression of the defining Chern polynomial.

Corollary 3.5. For every exact sequence of locally free sheaves $0 \to E' \to E \to E'' \to 0$,

$$c(E) = c(E')c(E'')$$

in $A^*(S)$.

3.8. Comparison with classical Chern classes

If X is locally of finite type over $\operatorname{Spec} \mathbb{C}$, the analytic space $X(\mathbb{C})$ carries the usual topological theory of Chern classes. For a continuous complex vector bundle V on $X(\mathbb{C})$ one has classes

$$c_i(V) \in H^{2i}(X(\mathbb{C}), \mathbb{Z}).$$

If E is an algebraic locally free sheaf on X, its analytification gives such a bundle, and reduction modulo n gives classes in

$$H^{2i}(X(\mathbb{C}), \mathbb{Z}/n\mathbb{Z}).$$

The comparison isomorphism between analytic and étalecohomology identifies these with the étaleChern classes just defined. In particular, the square

$$\begin{array}{ccc} H^1(X, \mathcal{O}_X^*) & \longrightarrow & H^1(X(\mathbb{C}), \mathcal{O}_{X_{\mathrm{an}}}^*) \\ \partial \downarrow & & \downarrow c_1 \\ H^2(X, \mu_n) & \longrightarrow & H^2(X(\mathbb{C}), \mathbb{Z}/n\mathbb{Z}) \end{array}$$

commutes. This follows from the two Kummer-type exact sequences: algebraically,

$$0 \to \mu_n \to \mathcal{O}_X^* \xrightarrow{n} \mathcal{O}_X^* \to 0,$$

and analytically,

$$0 \to \mathbb{Z} \to \mathcal{O}_{X_{\mathrm{an}}} \xrightarrow{\exp} \mathcal{O}_{X_{\mathrm{an}}}^* \to 0.$$

Passing to the projective limit over $n = \ell^r$ defines the ℓ-adic Chern classes

$$c_i(E) \in H^{2i}(X, \mathbb{Z}_\ell(i)),$$

for every prime ℓ invertible on X.

4. The self-intersection formula and applications

Let $i : Y \hookrightarrow X$ be a regular closed immersion of codimension d between smooth schemes, and let $N = N_{Y/X}$ be the normal bundle. The cohomological Gysin morphism

$$i_* : H^m(Y, M) \to H^{m+2d}(X, M(d))$$

is characterized by the fundamental class of Y in X and by compatibility with pullback, cup-product, and trace.

Self-intersection formula. For $x \in H^*(Y, M)$ one has

$$i^* i_*(x) = c_d(N) \cup x.$$

The proof reduces by deformation to the normal bundle. In the vector-bundle case the zero-section $s : S \to N$ has normal bundle N, and the identity follows from the computation of the top Chern class through the projective completion

$$\widehat{N} = \mathbb{P}(N \oplus \mathcal{O}_S).$$

The projective bundle formula and the explicit description of the class of the zero-section give the equality.

One application is the Euler–Poincare class. If X is projective over an algebraically closed field k, the diagonal class in $X \times X$ has a self-intersection whose degree is

$$\chi_\ell(X) = \sum_i (-1)^i \dim_{\mathbb{Q}_\ell} H^i(X, \mathbb{Q}_\ell).$$

When X is projective, this class is represented in the Chow ring by the self-intersection of the diagonal cycle. Hence it is independent of the auxiliary prime $\ell \neq \mathrm{char}(k)$; one may speak simply of the Euler–Poincare characteristic $\chi(X)$.

5. Flag schemes

Let E be a locally free $\mathcal{O}_S$-module of rank n and let

$$n = p_1 + \cdots + p_k$$

be a partition. The flag scheme $D = D(p_1, \ldots, p_k; E)$ represents filtrations of E with successive quotients of ranks p_i. It is built by iterating Grassmannian or projective bundle constructions. Consequently the pullback

$$A^*(S) \longrightarrow A^*(D)$$

is injective, and $A^*(D)$ is a finite free $A^*(S)$-module with an explicit basis in Schubert monomials in the Chern classes of the tautological quotients.

The exposé follows Grothendieck's treatment of Chern classes in the Chevalley seminar. The key lemma used in flag calculations says that, for an exact sequence

$$0 \to L \to E \to F \to 0$$

with L invertible, the induced sequence of symmetric algebras

$$0 \to L \otimes S(E) \to S(E) \to S(F) \to 0$$

is exact as a sequence of graded $\mathcal{O}_S$-modules. This identifies each step in the flag tower with a regular immersion defined by an invertible ideal, allowing the Gysin formula to be applied inductively.

6. Group schemes

The same formalism applies to smooth group schemes and their homogeneous spaces. If a smooth group scheme G acts transitively on a smooth scheme X with stabilizer H, the quotient geometry gives a comparison between the cohomology of X, of G, and of H. When the quotient is built from flag varieties or from projective bundles, the projective-bundle theorem and Chern-class formulas compute its cohomology ring.

The point for later use is not merely the calculation itself but the functorial behavior: characteristic classes of the associated vector bundles are compatible with change of group, restriction to subgroups, and passage to quotient varieties. These compatibilities feed into the trace and Lefschetz formulas where equivariance is present.

7. Complete intersections

Let X be a smooth projective scheme and let $Y \subset X$ be a smooth complete intersection cut out by a regular sequence, or more generally by a regular section of a locally free sheaf. The normal bundle of Y in X is then identified with the restriction of that locally free sheaf. The self-intersection formula gives

$$i^* i_*(1) = c_d(N_{Y/X}).$$

Lefschetz theorem 7.1. For a smooth hyperplane section or suitable complete intersection, the pullback on cohomology is an isomorphism in the usual stable range and is injective at the middle degree boundary.

The exposé treats this as the étaleanalogue of the classical Lefschetz theorem. It also records the expected stronger Lefschetz statements suggested by the transcendental theory. The point is that Chern classes and Gysin morphisms supply the algebraic incarnation of the topological cycle classes used in the classical proof.

In particular, the Euler–Poincare characteristic of complete intersections is computed by applying Gauss–Bonnet to the total Chern class of the tangent bundle, with the normal exact sequence

$$0 \to T_Y \to T_X|_Y \to N_{Y/X} \to 0$$

and the multiplicativity of total Chern classes.

8. Blown-up varieties

Let X be regular and let $Y \subset X$ be a regular closed subscheme of codimension d. Let

$$p : X' \to X$$

be the blow-up of X along Y, and let $Y' = p^{-1}(Y)$ be the exceptional divisor. Then

$$Y' \simeq \mathbb{P}(N_{Y/X}),$$

where $N_{Y/X}$ is the normal bundle. Let $q : Y' \to Y$ be the projection and let $j : Y' \hookrightarrow X'$ be the exceptional divisor immersion.

The cohomology of X' is determined by that of X and Y. The additive decomposition is

$$H^*(X') \simeq p^* H^*(X) \oplus \bigoplus_{i=1}^{d-1} j_*\big(q^* H^{*-2i}(Y)(-i)\, \xi^{i-1}\big),$$

where $\xi = c_1(\mathcal{O}_{Y'}(1))$. The proof follows from the localization exact sequence for the pair (X', Y'), the corresponding sequence for (X, Y), and the projective bundle formula for $Y' \to Y$.

The multiplicative structure is determined by three rules. First,

$$p^*(a)p^*(b) = p^*(ab).$$

Second, the projection formula gives

$$p^*(a)\, j_*(z) = j_*(j^* p^*(a)\, z) = j_*(q^* i^*(a)\, z).$$

Third, products of exceptional terms are reduced to the self-intersection formula for j:

$$j^* j_*(z) = c_1(\mathcal{O}_{Y'}(-1))\, z = -\xi z.$$

Together with the Chern polynomial relation of the projective bundle $\mathbb{P}(N_{Y/X})$, these rules give the full ring structure.

9. Chow ring of a blow-up and self-intersection in the Chow ring

The last part of the exposé repeats the preceding calculation inside the Chow ring. Let X be noetherian regular, let $Y \subset X$ be regular of codimension d, let X' be the blow-up of X along Y, and let Y' be the exceptional divisor. Again $Y' \simeq \mathbb{P}(N_{Y/X})$.

Theorem 9.1. With the preceding notation, the homomorphism

$$A^*(X) \oplus \bigoplus_{i=1}^{d-1} A^{*-i}(Y) \longrightarrow A^*(X')$$

which sends $(a, b_1, \ldots, b_{d-1})$ to

$$p^* a + \sum_{i=1}^{d-1} j_*(q^* b_i\, \xi^{i-1})$$

is an isomorphism of abelian groups.

The proof is the Chow-theoretic version of the cohomological calculation. It uses the deformation to the normal cone and the projective bundle formula in Chow groups. The key computational lemma is the following.

Lemma 9.1. Let S be smooth and quasi-projective over a separably closed field, let E be locally free of rank $r+1$, and let $p : P = \mathbb{P}(E) \to S$. Let F be defined by

$$0 \to F \to p^*E \to \mathcal{O}_P(1) \to 0,$$

and put $\xi = c_1(\mathcal{O}_P(1))$. Then, for $y \in C^*(S)$,

$$y = p_!\big(p^*y\, c_r(F)\big).$$

If $E = N \oplus \mathcal{O}_S$ and $s : S \to P$ is the zero-section in the projective completion $\widehat{N}$, then

$$s_!(y) = p^*y\, \xi^r, \qquad s^*s_!(y) = y\, c_r(N^\vee),$$

and for $z \in C^*(\widehat{N})$,

$$s^*(z) = p_!\big(z\, c_r(F)\big).$$

This lemma is a direct analogue of the cohomological computation in the normal bundle. It reduces the self-intersection formula to the projective bundle calculation and the behavior of the zero-section.

The self-intersection formula in the Chow ring. Let $i : Y \hookrightarrow X$ be a regular closed immersion of codimension d, with normal bundle N. For $y \in A^*(Y)$ one has

$$i^*i_*(y) = c_d(N)\, y$$

in $A^*(Y)$.

This is Mumford's Chow-theoretic self-intersection formula. It is obtained by applying the preceding blow-up computation to the deformation to the normal cone, where the assertion becomes the explicit calculation for the zero-section.

9.3. Useful relations

Inside the Chow ring of the blow-up, the exceptional divisor class and the Chern classes of the normal bundle satisfy the relation

$$\xi^d - c_1(N)\xi^{d-1} + \cdots + (-1)^d c_d(N) = 0$$

on $Y' = \mathbb{P}(N)$. The exceptional divisor satisfies

$$j^* j_*(1) = -\xi.$$

Together with the projection formula, these relations determine all products involving exceptional classes.

9.4. Key lemma

The key lemma asserts that the kernel and cokernel appearing in the comparison of $A^*(X')$ with the direct sum of $A^*(X)$ and the shifted $A^*(Y)$ are killed successively by projective bundle trace maps. More concretely, for the filtration by powers of the exceptional divisor, the associated graded pieces are the corresponding graded pieces of

$$A^*(Y')[1, \xi, \dots, \xi^{d-1}],$$

modulo the Chern polynomial. The projective bundle formula identifies these pieces with $A^*(Y)$.

9.5. End of the proof

The proof of the blow-up theorem follows by combining the key lemma with the localization exact sequences

$$A^*(Y) \to A^*(X) \to A^*(X \setminus Y) \to 0$$

and

$$A^*(Y') \to A^*(X') \to A^*(X' \setminus Y') \to 0,$$

noting that $X' \setminus Y'$ is canonically isomorphic to $X \setminus Y$. The resulting diagram has exact rows. The projective bundle formula computes the left vertical term, and the five lemma gives the decomposition.

9.6. The key formula

The fundamental identity may be written in the compact form

$$j_*\big(q^*(y)\xi^m\big) \cdot j_*\big(q^*(z)\xi^n\big) = -j_*\big(q^*(yz)\xi^{m+n+1}\big),$$

with the understanding that powers ξ^a for $a \geq d$ are reduced by the Chern polynomial of N. This formula is simply the projection formula and the equality $j^*j_*(1) = -\xi$.

9.8. Proof of Proposition 9.6

Proposition 9.6 is proved by applying the preceding formula repeatedly and using the fact that $q^* : A^*(Y) \to A^*(Y')$ is injective. The exceptional contribution is expressed in the basis $1, \xi, \ldots, \xi^{d-1}$ over $A^*(Y)$; uniqueness of the expression follows from the projective bundle theorem. The compatibility with pushforward to X comes from

$$p_*j_* = i_*q_*.$$

9.9. Structure theorem for the Chow group of the blow-up

The structure theorem states that $A^*(X')$ is the direct sum of the pullback of $A^*(X)$ and the exceptional summands generated by $j_*(q^*A^*(Y)\xi^i)$ for $0 \leq i \leq d-2$. Multiplication is described by the formulas above. Equivalently, $A^*(X')$ is the $A^*(X)$-algebra generated by the exceptional divisor class, with relations induced by the Chern polynomial of N and the restriction of cycles from X to Y.

9.10. Interpretation and Riemann–Roch

The final interpretation identifies the blow-up formula with the Riemann–Roch behavior of the exceptional divisor and the normal bundle. The Chern character and Todd class transform as dictated by the exact sequences

$$0 \to T_{X'} \to p^*T_X \to \text{exceptional quotient} \to 0$$

and

$$0 \to T_Y \to T_X|_Y \to N \to 0.$$

Thus the blow-up formula is compatible with the expected Riemann–Roch identity: pushforward of characteristic classes through p recovers the classes on X, while the exceptional summands account for the correction terms supported on Y.

Bibliography of Exposé VII

The sources cited include Grothendieck's theory of Chern classes, the Chevalley seminar on Chow rings and applications, Mumford's proof of the self-intersection formula in the Chow ring, and the Lefschetz theorems used in the cohomological applications.

Exposé VIII – Class Groups of Abelian and Triangulated Categories. Perfect Complexes

Class groups of abelian and triangulated categories. Perfect complexes.

by A. Grothendieck, written by I. Bucur

This exposé recalls, in a rapid form, the notions and results needed later concerning Grothendieck groups, pseudo-coherent complexes, and perfect complexes. A more complete treatment, with detailed proofs, is announced for the first exposés of the seminar on the Riemann–Roch theorem for schemes, later SGA 6.

1. Abelian categories

Let $\mathcal{C}$ be an abelian category and $\mathcal{D}$ a full subcategory of $\mathcal{C}$. A function f from $\mathcal{D}$ to an abelian group G is called *additive* if, for every exact sequence

$$0 \to E' \to E \to E'' \to 0$$

in $\mathcal{C}$ whose three terms belong to $\mathcal{D}$, one has

$$f(E) = f(E') + f(E'').$$

From the definition one immediately obtains $f(0) = 0$, invariance under isomorphism, and the corresponding formula for exact sequences viewed in the opposite category. These elementary properties are the universal motivation for the Grothendieck group.

2. The Grothendieck group of an abelian category

Let $\mathcal{C}$ be an abelian category. Let $F(\mathcal{C})$ be the free abelian group on isomorphism classes $[E]$ of objects of $\mathcal{C}$. The Grothendieck group $K(\mathcal{C})$ is the quotient of $F(\mathcal{C})$ by the subgroup generated by the elements

$$[E] - [E'] - [E'']$$

for all exact sequences $0 \to E' \to E \to E'' \to 0$. The canonical map

$$k_{\mathcal{C}} : \mathrm{Ob}\,\mathcal{C}/\simeq \longrightarrow K(\mathcal{C})$$

is universal for additive functions out of $\mathcal{C}$: if f is additive with values in an abelian group G, then there is a unique homomorphism $g : K(\mathcal{C}) \to G$ such that $f = g \circ k_{\mathcal{C}}$.

If $u : \mathcal{C} \to \mathcal{C}'$ is exact, then u induces a homomorphism

$$K(u) : K(\mathcal{C}) \to K(\mathcal{C}'),$$

characterized by

$$K(u)([E]) = [u(E)].$$

This is the functoriality of the Grothendieck group in the abelian setting.

3. Triangulated categories

Let $\mathcal{C}$ be a triangulated category. The Grothendieck group $K(\mathcal{C})$ is defined as the free abelian group on isomorphism classes of objects, modulo the relations

$$[X] - [Y] + [Z] = 0$$

for every distinguished triangle

$$X \to Y \to Z \to X[1].$$

Thus $[Y] = [X] + [Z]$. An exact functor of triangulated categories $u : \mathcal{C} \to \mathcal{C}'$ induces

$$K(u) : K(\mathcal{C}) \to K(\mathcal{C}')$$

by the same rule $[E] \mapsto [u(E)]$.

If $\mathcal{C} = \mathrm{D}^b(\mathcal{A})$ is the bounded derived category of an abelian category, the canonical functor from $\mathcal{A}$ to $\mathrm{D}^b(\mathcal{A})$ induces an isomorphism

$$K(\mathcal{A}) \simeq K(\mathrm{D}^b(\mathcal{A})).$$

The class of a bounded complex $K^\bullet$ is

$$[K^\bullet] = \sum_i (-1)^i [H^i(K^\bullet)],$$

or equivalently the alternating sum of its terms whenever this makes sense.

4. Thick subcategories and localization

A full subcategory $\mathcal{C}'$ of a triangulated category $\mathcal{C}$ is *thick* if it is triangulated and closed under direct summands: any direct factor in $\mathcal{C}$ of an object of $\mathcal{C}'$ again belongs to $\mathcal{C}'$. If $\mathcal{C}'$ is thick, the quotient category

$$\mathcal{C}/\mathcal{C}'$$

is triangulated and comes with an exact functor $Q : \mathcal{C} \to \mathcal{C}/\mathcal{C}'$ which is universal among exact functors from $\mathcal{C}$ annihilating $\mathcal{C}'$.

There is a natural exact sequence

$$K(\mathcal{C}') \longrightarrow K(\mathcal{C}) \xrightarrow{K(Q)} K(\mathcal{C}/\mathcal{C}') \longrightarrow 0.$$

This is the categorical form of localization. It will later be used to pass from complexes with torsion, or with prescribed support, to quotient categories where those complexes vanish.

5. Pseudo-coherent complexes

Let $\mathcal{C}$ be an abelian category and let $\mathcal{C}_0$ be a full subcategory satisfying stability conditions under kernels, cokernels, and the subquotients needed to perform homological constructions. We denote by

$$\mathrm{D}^{\pm}_{\mathcal{C}_0}(\mathcal{C})$$

the essential image of $\mathrm{D}^{\pm}(\mathcal{C}_0)$ in $\mathrm{D}(\mathcal{C})$. Its objects are called $\mathcal{C}_0$-pseudo-coherent complexes. Equivalently, a complex of $\mathcal{C}$ is $\mathcal{C}_0$-pseudo-coherent if it is isomorphic in $\mathrm{D}(\mathcal{C})$ to a complex bounded above whose terms lie in $\mathcal{C}_0$.

This subcategory is triangulated. Indeed, if two terms in a distinguished triangle

$$U \to V \to W \to U[1]$$

are pseudo-coherent, represent them by complexes $X^\bullet$ and $Y^\bullet$ with terms in $\mathcal{C}_0$. After replacing $X^\bullet$ by a quasi-isomorphic complex, the morphism is represented by an actual morphism of complexes. Its mapping cone has terms in $\mathcal{C}_0$ by the stability assumptions, and represents the third object W.

The intersection of $\mathrm{D}^{\pm}_{\mathcal{C}_0}(\mathcal{C})$ with $\mathrm{D}^b(\mathcal{C})$ is denoted

$$\mathrm{D}^{\pm,b}_{\mathcal{C}_0}(\mathcal{C}).$$

An object M of $\mathcal{C}$ is pseudo-coherent if it is pseudo-coherent as a complex concentrated in degree zero; equivalently, it admits a resolution on the left by objects of $\mathcal{C}_0$.

Proposition 5.1. A complex $X^\bullet$ is $\mathcal{C}_0$-pseudo-coherent if and only if it is isomorphic to a bounded-above complex whose terms are $\mathcal{C}_0$-pseudo-coherent objects.

The proof is deferred to SGA 6, where the theory is treated systematically.

6. Perfect complexes

An object of $\mathrm{D}(\mathcal{C})$ is called $\mathcal{C}_0$-perfect if it is isomorphic to an object of $\mathrm{D}^b(\mathcal{C}_0)$, that is, to a bounded complex all of whose terms lie in $\mathcal{C}_0$. The full subcategory of perfect complexes is denoted

$$\mathrm{D}^{\mathrm{parf}}_{\mathcal{C}_0}(\mathcal{C}).$$

It is the essential image of $\mathrm{D}^b(\mathcal{C}_0)$ in $\mathrm{D}(\mathcal{C})$. It is a triangulated subcategory, stable under direct summands under the standard hypotheses. The Grothendieck group of perfect complexes is therefore defined by the preceding triangulated construction.

When $\mathcal{C}$ is a category of modules and $\mathcal{C}_0$ is the category of finite free or finite projective modules, this recovers the usual perfect complexes. In geometry, perfect complexes are precisely the complexes locally quasi-isomorphic to bounded complexes of finite locally free modules.

7. An important special case

Let A be a ring and let $\mathcal{C}$ be the category of A-modules. Let $\mathcal{C}_0$ be the category of finite projective A-modules. Then a complex is perfect if and only if it is locally, for the relevant topology, quasi-isomorphic to a bounded complex of finite projective modules. The resulting Grothendieck group is the K-group used in Riemann–Roch formulas.

This special case is central because cohomology complexes with constructible coefficients often become perfect after imposing finite Tor-dimension conditions. In that situation, the trace of an endomorphism is computed in the Grothendieck group and then evaluated by a rank, character, or local term.

8. Tor of complexes

Let $\mathcal{C}_\ell$ denote a category of modules over a coefficient ring such as $\mathbb{Z}_\ell$ or $\mathbb{Z}/\ell^n\mathbb{Z}$. A complex X of $\mathrm{D}(\mathcal{C}_\ell)$ has *finite Tor-dimension* if it is bounded and if there exists an integer n_0 such that

$$\mathrm{Tor}^i(X, M) = 0$$

for every object M and every $i > n_0$ after the usual shift convention. Equivalently, tensoring with X has uniformly bounded cohomological amplitude.

Finite Tor-dimension is the condition which permits one to define derived tensor products without leaving bounded categories. It is also the condition which makes duality behave as expected: a perfect complex has finite Tor-dimension, and its dual is again perfect.

9. Reminders on linear representations of finite groups

Let G be a finite group and A a commutative ring. The category of finite projective $A[G]$-modules has a Grothendieck group $K(A[G])$. If A is a field of characteristic not dividing $|G|$, this group is controlled by characters. In the integral ℓ-adic setting one uses projective $\mathbb{Z}_\ell[G]$-modules and compares them after extending scalars to $\mathbb{Q}_\ell$.

Induction and restriction are exact functors and therefore induce maps on Grothendieck groups:

$$\mathrm{Ind}_H^G : K(A[H]) \to K(A[G]), \qquad \mathrm{Res}_H^G : K(A[G]) \to K(A[H]).$$

Duality sends a module M to

$$M^* = \mathrm{Hom}_A(M, A),$$

with the contragredient action. If the character of M is χ, then that of M^* is $g \mapsto \chi(g^{-1})$.

4. The Swan representation

The next part of the source turns to the local representation attached to a ramified Galois cover of curves. Let C be a smooth, proper, connected curve over an algebraically closed field k, with generic point η and function field $K = k(\eta)$. Let

$$K \subset K', \qquad [K' : K] = n$$

be a finite Galois extension with group G, and let C' be the normalization of C in K'. The canonical map is

$$p : C' \to C.$$

The group G acts on C'. For $y' \in C'$ write $G_{y'}$ for its stabilizer.

If Π is a uniformizer at y' and $s \in G_{y'}$, $s \neq e$, define

$$v_{y'}(s) = v_{y'}(s(\Pi) - \Pi).$$

This is the multiplicity of y' as a fixed point of the automorphism $s : C' \to C'$. The equality follows from the local intersection calculation: the diagonal and the graph of s meet with multiplicity $v_{y'}(s(\Pi) - \Pi)$.

4.2. Definition of the Swan representation

The Swan character attached to y' is the class function

$$\mathrm{sw}_{y'}(s) = \begin{cases} -1 - v_{y'}(s), & s \neq e, \\ \sum_{s\neq e} v_{y'}(s), & s = e. \end{cases}$$

The theorem of Artin–Serre–Swan asserts that, for a prime $\ell \neq \mathrm{char}(k)$, there is a finite projective $\mathbb{Z}_\ell[G_{y'}]$-module $\mathrm{Sw}_{y'}$, unique up to isomorphism, having this character.

One also considers

$$\widetilde{\mathrm{Sw}}_{y'} = \mathrm{Sw}_{y'} \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

Its character is

$$\widetilde{\mathrm{sw}}_{y'}(s) = \begin{cases} -1, & s \neq e, \\ |G_{y'}| - 1, & s = e. \end{cases}$$

Finally, the Artin character is

$$\mathrm{art}_{y'} = \mathrm{sw}_{y'} + \mathbf{1}_{G_{y'}},$$

where $\mathbf{1}_{G_{y'}}$ denotes the character of the augmentation representation. Explicitly,

$$\mathrm{art}_{y'}(s) = \begin{cases} -v_{y'}(s), & s \neq e, \\ \sum_{s\neq e} v_{y'}(s) + |G_{y'}|, & s = e. \end{cases}$$

Let $y = p(y')$. By induction from $G_{y'}$ to G set

$$\begin{aligned} \mathrm{Sw}_y &= \mathrm{Sw}_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G], \\ \widetilde{\mathrm{Sw}}_y &= \widetilde{\mathrm{Sw}}_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G], \\ \mathrm{Art}_y &= \mathrm{Art}_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G]. \end{aligned}$$

Their characters are denoted sw_y, $\widetilde{\mathrm{sw}}_y$, and art_y.

4.3. Independence of the point above y

The induced modules just defined depend, up to isomorphism, only on $y \in C$, not on the chosen point y' above y. If $y'' = gy'$, then the situation at y'' is obtained from that at y' by the automorphism g of C' and by conjugation $s \mapsto gsg^{-1}$ on G. Extension of operators through this inner automorphism sends the corresponding module to an isomorphic one. Hence Sw_y, $\widetilde{\mathrm{Sw}}_y$, and Art_y are well-defined.

Proposition 4.4. With the preceding notation, and for $M^\bullet \in \mathrm{D}^*(\mathbb{Z}_\ell[G])$, one has canonical isomorphisms

$$\mathrm{Sw}_{y'}^* = \mathrm{Sw}_{y'},$$

and

$$\mathrm{Hom}_{\mathbb{Z}_\ell[G]}(\mathrm{Sw}_{y'}, M^\bullet) \simeq \mathrm{Sw}_{y'} \otimes_{\mathbb{Z}_\ell[G]} M^\bullet.$$

Proof. Finite projective $\mathbb{Z}_\ell[G]$-modules are determined by their characters in this setting. If σ is the character of $\mathrm{Sw}_{y'}$, then the dual has character $\sigma^*(s) = \sigma(s^{-1})$. But

$$v_{y'}(s^{-1}(\Pi) - \Pi) = v_{y'}(s^{-1}(s(\Pi) - \Pi)) = v_{y'}(s(\Pi) - \Pi),$$

so the Swan character is invariant under inversion. This gives self-duality; the Hom–tensor formula follows.

4.5. Explicit character formulas

For later use the character of the induced Swan representation can be written explicitly. For $s \in G$,

$$\mathrm{sw}_y(s) = \sum_{p(y')=y} \mathrm{sw}_{y'}(s),$$

with terms understood to be zero unless s fixes the corresponding point. Thus

$$\mathrm{sw}_y(s) = \begin{cases} \sum_{p(y')=y}(1 - v_{y'}(s)), & s \neq e, \\ 0, & s = e, \end{cases}$$

under the convention of the source. Similarly

$$\mathrm{art}_y(s) = \begin{cases} \sum_{p(y')=y}(-v_{y'}(s)), & s \neq e, \\ 0, & s = e, \end{cases}$$

and

$$\widetilde{\mathrm{sw}}_y(s) = \begin{cases} \sum_{p(y')=y}(-1), & s \neq e, \\ 0, & s = e. \end{cases}$$

These formulas are obtained from the character formula for induced representations, choosing representatives for the right cosets of $G_{y'}$ in G.

4.6. The non-connected case

The invariants Sw_y, $\widetilde{\mathrm{Sw}}_y$, and Art_y are defined under slightly more general hypotheses. Let C' be a smooth algebraic curve over k, not necessarily connected, with an action of a finite group G. Suppose there is a dense open subset U of $C = C'/G$ such that $C'|_U$ is a principal G-cover of U.

For a point $y \in C$, choose $y' \in C'$ above y. Let $C'_{y'}$ be the irreducible component of C' containing y', and C_y the component of C containing y. Let G'_y be the stabilizer of the component $C'_{y'}$. Then the extension of function fields

$$k(\eta_y) \subset k(\eta'_y)$$

is Galois with group G'_y, and $C'_{y'}$ is the normalization of C_y in that extension. Applying the connected construction to $C'_{y'} \to C_y$ gives projective $\mathbb{Z}_\ell[G'_y]$-modules. The desired $\mathbb{Z}_\ell[G]$-modules Sw_y, $\widetilde{\mathrm{Sw}}_y$, and Art_y are their inductions from G'_y to G.

Remark 4.6. The modules Sw_y, $\widetilde{\mathrm{Sw}}_y$, and Art_y are purely local invariants. They are determined by the action of the stabilizer on the completed local ring $\widehat{\mathcal{O}}_{C',y'}$. Moreover $\mathrm{Sw}_y = 0$ if and only if the cover is tamely ramified at y'; equivalently, the stabilizer $G_{y'}$ has order prime to $\mathrm{char}(k)$.

Exposé XI (end) – Euler–Poincare Characteristic and Weil Formulas

5. The Weil formula

This final part of Exposé XI compares the trace formalism with the classical formula of Weil. The situation is that of a smooth proper curve C over an algebraically closed field, an open dense subset $U \subset C$, a finite Galois covering $p : C' \to C$ with group G, and a coefficient ring Λ such as $\mathbb{Z}_\ell$ or a finite quotient. The sheaves under consideration are constructible on C, locally constant on U, and controlled near the boundary by the local Galois data of the covering.

The Euler–Poincare characteristic is written

$$\chi(C, F) = \sum_i (-1)^i [H^i(C, F)]$$

in the Grothendieck group of Λ-modules, or in the corresponding Grothendieck group of $\Lambda[G]$-modules when an equivariant structure is present. The goal is to express the difference between the global Euler characteristic and the rank contribution in terms of local ramification classes.

Proposition 5.1. Let F be a constructible sheaf on C, locally constant on U, and suppose that its pullback to U' is represented by a finite projective $\Lambda[G]$-module. Then the equivariant Euler characteristic of F is the sum of the unramified global term and local correction terms at the points of $C \setminus U$.

The proof is a formal consequence of excision and induction. One chooses a finite Galois cover which trivializes the sheaf on U, writes the cohomology on U in terms of the cohomology upstairs together with the G-module of fibers, and compares compactly supported cohomology with cohomology on the compact curve. The boundary contributions are precisely the Swan and Artin representations defined in Exposé VIII.

6. Definition of the local terms $\varepsilon_x^\Delta(F)$

Let $x \in C \setminus U$. The local term $\varepsilon_x^\Delta(F)$ is an element of a Grothendieck group, attached to the action of the decomposition group at a point x' above x. It measures the defect between the representation carried by the generic fiber of F and the representation seen after extending across x. In concrete terms it is built from the Swan representation and the fiber $F_{\bar{\eta}}$:

$$\varepsilon_x^\Delta(F) = [\mathrm{Sw}_x \otimes F_{\bar{\eta}}] + \text{tame correction} - \text{invariant correction}.$$

The precise expression depends on whether one writes the formula in $K(\Lambda)$, in $K(\Lambda[G])$, or after applying a character. The important point is that the local term is determined by the completed local ring and the local monodromy representation.

The notation Δ emphasizes that this term is the local contribution at the diagonal in the Lefschetz trace formula: it is the same obstruction which appears when one computes the intersection of the graph of an endomorphism with the diagonal.

7. Euler–Poincare formula

Theorem 7.1. Let C be a smooth proper connected curve over an algebraically closed field, let $U \subset C$ be a dense open subset, and let F be a constructible Λ-sheaf on C, locally constant on U. Then

$$\chi(C, F) = \mathrm{rk}(F)\chi(C, \Lambda) - \sum_{x \in C \setminus U} a_x(F),$$

where $a_x(F)$ is the Artin conductor, expressed in K-theoretic form by the local term $\varepsilon_x^{\Delta}(F)$.

Proof. The proof proceeds in four steps.

First, for a bounded-below complex Ψ of sheaves of $\Lambda[G]$-modules on C, one uses the projection formula and the exactness of induction to identify the equivariant Euler characteristic of $p_*\Psi$ with the induced class from the cohomology of Ψ upstairs.

Second, if Φ is a bounded-below complex on C which is zero outside U, its compactly supported cohomology is computed by extension by zero and the preceding equivariant comparison.

Third, the local boundary triangle

$$j_!j^*F \to F \to i_*i^*F \to$$

expresses $\chi(C, F)$ as a sum of the compactly supported contribution over U and the stalk contributions at $C \setminus U$.

Finally, by choosing a cover $U' \to U$ which trivializes the sheaf and extending to a finite cover $C' \to C$, the local terms are identified with the Swan and Artin modules. This yields the stated conductor formula.

Corollary 7.9. If F is locally constant on all of C, then all local conductor terms vanish and

$$\chi(C, F) = \mathrm{rk}(F)\chi(C, \Lambda).$$

Corollary 7.12. For a constructible sheaf F on a smooth curve, the Euler characteristic depends only on the generic rank, the Euler characteristic of the curve, and the local ramification invariants at the finitely many points where F is not locally constant.

Lemma 7.14. The local conductor class is additive in distinguished triangles and in short exact sequences of constructible sheaves. It is invariant under replacing a local trivializing cover by a dominating cover.

The proof is a diagram chase in the Grothendieck group, using exactness of restriction, induction, and the additivity of the Swan representation. This is the form which makes the Euler–Poincare formula independent of auxiliary choices.

References for Exposé XI

The references are to the preceding theory of local representations, to the Grothendieck–Ogg–Shafarevich formula, and to the trace formula for sheaves on curves.

Exposé XII – Nielsen–Wecken and Lefschetz Formulas in Algebraic Geometry

Nielsen–Wecken and Lefschetz formulas in algebraic geometry

1. General setting

The exposé studies fixed point formulas for correspondences and endomorphisms in algebraic geometry, with particular attention to local terms. The input consists of a scheme X, an endomorphism or correspondence f, a constructible sheaf or complex F, and sometimes a finite group G acting on a covering $X' \to X$. One wants a formula expressing the trace of the induced endomorphism on cohomology as a sum over fixed point classes.

In topology this is the Nielsen–Wecken refinement of the Lefschetz fixed point formula. In algebraic geometry, the role of a fixed point class is played by a connected component of the intersection of the graph of the correspondence with the diagonal, or by its lift to a finite Galois covering.

2. Fixed points and lifted fixed points

Let $f : X \to X$ be an endomorphism and let $p : X' \to X$ be a finite covering with group G. A lift $f' : X' \to X'$ need not commute with the G-action; it may instead satisfy

$$f'g = \varphi(g)f'$$

for an endomorphism $\varphi : G \to G$. For $g \in G$ one considers the fixed points of

$$g^{-1}f' : X' \to X'.$$

The local terms attached to these fixed points are then assembled into a class function on G, or into an element of a Grothendieck group after applying traces.

This is the algebraic form of separating fixed points into Nielsen classes. The summation over $g \in G$ recovers the ordinary Lefschetz number downstairs.

3. The local Nielsen–Wecken invariant

Let x be a fixed point downstairs, and choose points x' above it. The local invariant depends on the intersection multiplicity of the graph of $g^{-1}f'$ with the diagonal at x', and on the trace of the induced endomorphism on the stalk of the sheaf. In the case of curves, if R is the completed local ring, π a uniformizer, and h the induced endomorphism, the fixed point multiplicity is

$$\nu(h) = v(h(\pi) - \pi).$$

This is the same local intersection calculation used earlier for the Swan representation.

4. Generalities on local terms

For a prime $\ell \neq \mathrm{char}(k)$, the local term associated with x is denoted

$$S_{x,\ell}(f', f, G, \varphi).$$

It is a rational expression obtained by summing the local traces over the lifted fixed points.

Conjecture 4.5. For every $\ell \neq \mathrm{char}(k)$, the values of $S_{x,\ell}(f', f, G, \varphi)$ are ℓ-adic integers.

The exposé proves this conjecture in dimension one by reducing the assertion to an explicit calculation in complete discrete valuation rings.

Proposition 4.6. Let R be a discrete valuation ring with maximal ideal $\mathfrak{m}$, uniformizer π, algebraically closed residue field k, and valuation v. Let $f'_R : R \to R$ be an endomorphism and let G be a finite group of automorphisms of R. Assume that f'_R and all elements of G induce the identity on k, and that an endomorphism $\varphi : G \to G$ is given with the compatibility

$$f'_R \circ g = \varphi(g) \circ f'_R.$$

Then the local expression defining the Nielsen–Wecken term is an ℓ-adic integer.

The proof reduces to the case $g = e$ and $\varphi = \mathrm{id}$, by factoring through the kernel on which the action is relevant. Let

$$r = \nu(f'_R) = v(f'_R(\pi) - \pi).$$

If $r \geq 2$, the following lemma controls the valuations of the iterates and of the terms obtained after applying elements of G.

Lemma 4.7. Under the hypotheses of Proposition 4.6, assume in addition that $\varphi = \mathrm{id}_G$ and $r \geq 2$. Then the valuation r satisfies the stability conditions required to compute the local fixed point multiplicities, and the resulting local term is integral.

The verification is a direct calculation in R. One writes $f'_R(\pi) = \pi + u\pi^r + \cdots$ with u a unit and tracks the effect of automorphisms $g \in G$ on the leading term. The compatibility with f'_R forces the same leading valuation for the relevant differences. Thus the denominator which appears in the formal local term is cancelled by the multiplicity.

Corollary 4.8. The rational local numbers obtained from Proposition 4.6 have ℓ-adic integral images for every $\ell \neq \mathrm{char}(k)$.

Proposition 4.9. Conjecture 4.5 is true when $\dim X = 1$.

The proof introduces the endomorphism on the completed local rings at the fixed points, reduces to the complete DVR calculation, and sums over the finite set of lifted fixed points.

5. Generalization of the Nielsen–Wecken formula

Theorem 5.1. In the preceding equivariant situation, the trace of the induced endomorphism on the cohomology of X is equal to the sum of the local Nielsen–Wecken terms over the fixed point classes:

$$\mathrm{Tr}(f^* \mid R\Gamma(X, F)) = \sum_x S_{x,\ell}(f', f, G, \varphi; F).$$

Proof. One first reduces modulo ℓ^n for every $n \geq 1$, where the finite coefficient Lefschetz formula is available. The equivariant decomposition of the covering separates the trace into contributions indexed by $g \in G$. Passing to the inverse limit and using the integrality proved in dimension one gives the asserted ℓ-adic formula. Compatibility of the local terms with reduction modulo ℓ^n ensures that the resulting expression is independent of n.

5.10. Special cases

If the covering is unramified near the relevant fixed points, the Swan terms vanish and the local formula reduces to the ordinary local Lefschetz multiplicity. If all multiplicities

$$\nu_{x'}(g^{-1}f')$$

are equal to 1 at the lifted fixed points, then each local contribution is simply the trace on the corresponding stalk, weighted by the number of lifted fixed points in the Nielsen class.

6. Application to a Lefschetz formula

Let X be proper and let $f : X \to X$ be an endomorphism with isolated fixed points, or more generally with a fixed point scheme satisfying the usual cleanliness hypotheses. Let F be a constructible complex with an endomorphism compatible with f. The Lefschetz formula reads

$$\sum_i (-1)^i \operatorname{Tr}(f^* \mid H^i(X, F)) = \sum_{x \in \operatorname{Fix}(f)} LT_x(f, F).$$

The local term $LT_x(f, F)$ is computed by the Nielsen–Wecken construction after choosing a suitable covering and summing over the lifts.

6.4. Remarks on local terms

If $g \in G$, conjugation by g transports the fixed point data at x' to that at gx', and the local terms transform accordingly. Thus the local contribution depends only on the conjugacy class of the local stabilizer data.

Let x' be a fixed point of a lift and let $G_{x'}$ be its stabilizer. This stabilizer is stable under φ, and the local term may be computed entirely in the complete local ring at x' together with the induced endomorphism of $G_{x'}$. In the tamely ramified case, the Swan contribution vanishes; the local term is then the tame trace multiplied by the geometric fixed point multiplicity.

Lemma 6.4.5. If the ramification at the fixed point is tame and the fixed point multiplicity is simple, the local term is the ordinary trace of the induced endomorphism on the stalk.

When the sheaf F is unramified at all fixed points, the formula simplifies to the standard Lefschetz trace formula with the local terms given by stalk traces and intersection multiplicities.

7. Comments on validity hypotheses

The final comments emphasize that the formula depends on finiteness, properness, and constructibility assumptions. The fixed point locus must be proper over the base, or else cohomology with proper supports must be used. The sheaf must have finite Tor-amplitude in the relevant coefficient category, so that traces are defined. When these hypotheses hold, the local terms are stable under finite coefficient reduction and hence pass to the ℓ-adic limit.

7. Complements: end of Exposé XII

The end of the exposé adds compatibility statements for the local terms. They are invariant under replacing the Galois cover by a dominating cover, additive in distinguished triangles of coefficients, and compatible with the specialization maps used in the proof of the curve case. These facts allow the formula to be applied without making the auxiliary covering part of the final data.

The bibliography of Exposé XII records the topological Nielsen–Wecken sources, the Grothendieck–Verdier trace formalism, and the preceding SGA exposés on local terms and Euler–Poincare formulas.

Exposé XV – Frobenius Morphism and Rationality of L-Functions

Frobenius morphism and rationality of L-functions

1. Frobenius morphism

1.1. Frobenius endomorphism

Let X be a scheme of characteristic $p > 0$. The absolute Frobenius endomorphism

$$F_X : X \to X$$

is the identity on the underlying topological space and raises functions to their pth powers:

$$a \longmapsto a^p.$$

If X is defined over $\mathbb{F}_q$, one also considers the qth Frobenius, obtained by iterating F_X so that functions are sent to a^q.

For a geometric point $\bar{x}$ over a closed point $x \in |X|$, the Frobenius element acts on the geometric stalk of a constructible sheaf. The convention in the trace formula is to use geometric Frobenius, inverse to the arithmetic Frobenius on residue fields.

1.2. Relative Frobenius

Let $X \to S$ be a morphism of schemes in characteristic p. The absolute Frobenius maps of X and S form a commutative square only after twisting the structure morphism. The relative Frobenius is the morphism

$$F_{X/S} : X \to X^{(p)} = X \times_{S,F_S} S.$$

When $S = \operatorname{Spec} k$ for a perfect field k, the Frobenius twist $X^{(p)}$ is another k-scheme, often identified with X after choosing the Frobenius automorphism of k.

The relative Frobenius is functorial in X/S, compatible with products, and finite under the usual finite type hypotheses. It is the geometric morphism which relates point-counting over finite fields to traces on cohomology.

1.3. Effect on a module

If M is a module over a ring A of characteristic p, the Frobenius pullback is

$$F_A^* M = A \otimes_{F_A, A} M.$$

For sheaves this construction is local on X. A semilinear Frobenius structure on a sheaf is an isomorphism between the pullback by Frobenius and the sheaf itself, or dually an action of Frobenius on the stalks. Such structures are the coefficient data in the cohomological interpretation of L-functions.

1.4. Frobenius and cohomology

Let X be separated of finite type over $\mathbb{F}_q$ and let F be a constructible ℓ-adic sheaf. The geometric Frobenius acts on

$$H_c^i(X_{\overline{\mathbb{F}}_q}, F).$$

The Lefschetz trace formula gives

$$\sum_i (-1)^i \operatorname{Tr}\Big(F^n \mid H_c^i(X_{\overline{\mathbb{F}}_q}, F)\Big) = \sum_{x \in X(\mathbb{F}_{q^n})} \operatorname{Tr}\Big(F_x^n \mid F_{\bar{x}}\Big).$$

This identity is the bridge between Euler products and determinants of Frobenius on cohomology.

2. Rationality of L-functions

2.1. The theorem

Let X be separated of finite type over $\mathbb{F}_q$ and let F be a constructible $\mathbb{Q}_\ell$-sheaf on X. Define

$$L(X, F; t) = \prod_{x \in |X|} \det(1 - F_x t^{\deg x} \mid F_{\bar{x}})^{-1}.$$

Theorem. The function $L(X, F; t)$ is rational. More precisely,

$$L(X, F; t) = \prod_i \det(1 - Ft \mid H^i_c(X_{\overline{\mathbb{F}}_q}, F))^{(-1)^{i+1}}.$$

This is Grothendieck's cohomological rationality theorem for L-functions. The proof is based on the trace formula and on a reduction which permits one to apply the formula to enough powers of Frobenius.

2.2. A reduction lemma

The reduction lemma says that it suffices to prove the theorem after replacing X by a stratification on which F is locally constant and after replacing F by a sheaf trivialized by a finite cover. Additivity of compactly supported cohomology and multiplicativity of Euler products in distinguished triangles reduce the problem to these cases.

One also reduces to curves in the key step. Given a closed point contribution, one chooses curves passing through the point and uses the Lefschetz formula on those curves. The local computations of Exposés XI and XII control the terms which arise in this passage.

2.3. Proof of the theorem

The logarithm of the Euler product is

$$\log L(X, F; t) = \sum_{n \geq 1} \frac{t^n}{n} \sum_{x \in X(\mathbb{F}_{q^n})} \operatorname{Tr}(F_x^n \mid F_{\bar{x}}).$$

By the Lefschetz trace formula the inner sum is

$$\sum_i (-1)^i \operatorname{Tr}(F^n \mid H_c^i(X_{\overline{\mathbb{F}}_q}, F)).$$

For a finite-dimensional vector space V with endomorphism T one has

$$\exp\left(\sum_{n \geq 1} \operatorname{Tr}(T^n) \frac{t^n}{n}\right) = \det(1 - Tt)^{-1}.$$

Applying this identity to each cohomology group gives the determinant formula, hence rationality.

First reduction

The first reduction is from arbitrary constructible sheaves to sheaves which are locally constant on a dense open subset and zero or controlled on its complement. This is done by noetherian induction on the support and by the localization triangle

$$j_! j^* F \to F \to i_* i^* F \to .$$

Second reduction

The second reduction passes to a finite Galois cover which trivializes the locally constant sheaf. The equivariant form of the trace formula expresses the downstairs trace as an average of traces upstairs, weighted by the relevant group elements. This is where the Nielsen–Wecken and local term formulas enter.

Third reduction

The third reduction handles ramification. By compactifying and using the curve case, local ramification terms are shown to contribute rational factors. The integrality and invariance of the local terms ensure that these factors are independent of auxiliary choices.

End of the proof

After the reductions, the trace formula applies directly. The determinant expression obtained from compactly supported cohomology is a rational function, and the coefficient comparison of logarithmic derivatives identifies it with the Euler product. Thus the L-function is rational.

Bibliography of Exposé XV

The bibliography includes Grothendieck's Bourbaki exposé on the Lefschetz formula and rationality of L-functions, the preceding SGA trace-formula exposés, and the finiteness and constructibility theorems for étalecohomology.

Terminological index

The source closes with a terminological index. In modern English usage the principal entries in this range are: Artin conductor, constructible ℓ-adic sheaf, Euler–Poincare characteristic, Frobenius morphism, geometric Frobenius, Grothendieck group, Lefschetz local term, Nielsen–Wecken class, perfect complex, pseudo-coherent complex, relative Frobenius, Swan representation, and trace formula.

Index of notation

The main notation in this range includes F_X for absolute Frobenius, $F_{X/S}$ for relative Frobenius, H_c^i for compactly supported cohomology, $K(\mathcal{C})$ for a Grothendieck group, Sw_y and Art_y for local Swan and Artin representations, $Z(X, F; t)$ and $L(X, F; t)$ for zeta and L-functions, and $\mathrm{D}^{\mathrm{parf}}$ for perfect complexes.